Name

Mathematics Advanced Year 11

Trigonometry with Angles of any Magnitude

Book 1

Version: 261021

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Syllabus Content

MAV-11-04 applies trigonometry to solve problems involving geometric shapes

Trigonometry with angles of any magnitude

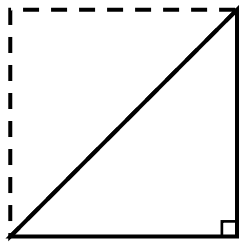
- Represent angles of any magnitude using rays from the origin in the Cartesian plane and describe how one ray represents infinitely many angles
- Develop the definitions $\cos\theta = \frac{x}{r}$, $\sin\theta = \frac{y}{r}$ and $\tan\theta = \frac{y}{x}$, where (x, y) is a point on the circle of radius r, centred at the origin, and θ is the angle between the positive x-axis and the radius drawn to this point
- Use the definitions $\cos\theta = \frac{x}{r}$, $\sin\theta = \frac{y}{r}$ and $\tan\theta = \frac{y}{x}$ to evaluate $\sin\theta$, $\cos\theta$ and $\tan\theta$ where θ is a multiple of $90°$, and identify the values of θ for which these ratios are undefined
- Extend and apply the definitions $\cos\theta = \frac{x}{r}$, $\sin\theta = \frac{y}{r}$ and $\tan\theta = \frac{y}{x}$ for angles of any magnitude
- Identify the related angle of an angle of any magnitude, excluding multiples of $90°$, as the acute angle between the ray and the x-axis and obtain the values of the trigonometric functions of an angle of any magnitude from the trigonometric functions of the related angle
- Develop and use the trigonometric ratios for angles that can be written in the form $\theta = 180° \pm A$, and $\theta = 360° - A$, where $0° < A < 90°$
- Establish and use the results $\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$ and $\tan(-\theta) = -\tan\theta$
- Examine the proof of the sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, cosine rule $c^2 = a^2 + b^2 - 2ab\cos C$ and the area of a triangle formula $A = \frac{1}{2}ab\sin C$ for a given triangle ABC
- Use graphing applications or geometric construction to examine the ambiguous case of the sine rule, in which there are two possible solutions for an angle, and the condition for it to arise
- Apply the sine rule, cosine rule and formula for the area of a triangle to solve problems where angles are measured in degrees, or degrees and minutes

Exact Values of Trigonometric Ratios (Acute Angles)

Investigation Exact values for 45°

Consider a unit square cut in half along the diagonal.

a Label the sides and angles of the triangle.



b Determine the exact values of:

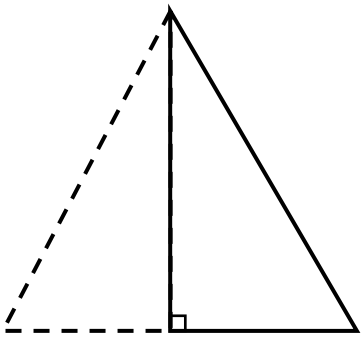
$\sin 45° =$

$\cos 45° =$

$\tan 45° =$

Consider a unit equilateral triangle cut in half along the altitude.

a Label the sides and angles of the triangle.



b Determine the exact values of:

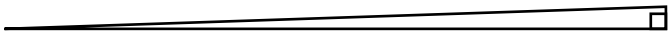
$\sin 30° =$ $\sin 60° =$

$\cos 30° =$ $\cos 60° =$

$\tan 30° =$ $\tan 60° =$

Consider a right-angled triangle with the two non-right angles $= 0°$ and $= 90°$.

⚠ A triangle doesn't exist when an angle is $0°$, we are finding the values of the ratios.



Determine the exact values of:

$\sin 0° =$ $\sin 90° =$

$\cos 0° =$ $\cos 90° =$

$\tan 0° =$ $\tan 90° =$

ⓘ Exact Values of the Trigonometric Ratios (0° 30° 45° 60° 90°)

	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

⚠ You must memorise these exact values and be able to recall them quickly.

1 Complete the table.

	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

2 State the exact value of the following.

a	$\cos 60°$ $\frac{1}{2}$	**b**	$\tan 30°$ $\frac{1}{\sqrt{3}}$	**c**	$\sin 60°$ $\frac{\sqrt{3}}{2}$	**d**	$\cos 30°$ $\frac{\sqrt{3}}{2}$
e	$\tan 60°$ $\sqrt{3}$	**f**	$\sin 30°$ $\frac{1}{2}$	**g**	$\cos 60°$ $\frac{1}{2}$	**h**	$\tan 30°$ $\frac{1}{\sqrt{3}}$
i	$\sin 30°$ $\frac{1}{2}$	**j**	$\cos 30°$ $\frac{\sqrt{3}}{2}$	**k**	$\tan 60°$ $\sqrt{3}$	**l**	$\sin 60°$ $\frac{\sqrt{3}}{2}$
m	$\cos 60°$ $\frac{1}{2}$	**n**	$\tan 30°$ $\frac{1}{\sqrt{3}}$	**o**	$\sin 30°$ $\frac{1}{2}$	**p**	$\cos 30°$ $\frac{\sqrt{3}}{2}$

3 State the exact value of the following.

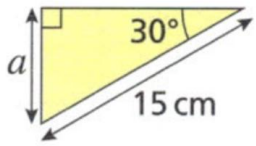
a	$\cos 45°$ $\frac{1}{\sqrt{2}}$	**b**	$\tan 0°$ 0	**c**	$\sin 90°$ 1	**d**	$\cos 0°$ 1
e	$\tan 45°$ 1	**f**	$\sin 45°$ $\frac{1}{\sqrt{2}}$	**g**	$\cos 90°$ 0	**h**	$\tan 90°$ 0
i	$\sin 0°$ 0	**j**	$\tan 0°$ 0	**k**	$\cos 45°$ $\frac{1}{\sqrt{2}}$	**l**	$\sin 45°$ $\frac{1}{\sqrt{2}}$
m	$\tan 45°$ 1	**n**	$\sin 90°$ 1	**o**	$\cos 0°$ 1	**p**	$\cos 45°$ $\frac{1}{\sqrt{2}}$

4 State the exact value of the following.

a	$\sin 0°$	0	**b**	$\cos 30°$	$\frac{\sqrt{3}}{2}$	**c**	$\sin 45°$	$\frac{1}{\sqrt{2}}$	**d**	$\cos 60°$	$\frac{1}{2}$
e	$\sin 30°$	$\frac{1}{2}$	**f**	$\tan 90°$	undefined	**g**	$\sin 90°$	1	**h**	$\sin 30°$	$\frac{1}{2}$
i	$\tan 60°$	$\sqrt{3}$	**j**	$\cos 45°$	$\frac{1}{\sqrt{2}}$	**k**	$\cos 60°$	$\frac{1}{2}$	**l**	$\tan 0°$	1
m	$\tan 0°$	0	**n**	$\cos 0°$	1	**o**	$\cos 90°$	0	**p**	$\cos 90°$	0
q	$\tan 30°$	$\frac{1}{\sqrt{3}}$	**r**	$\tan 45°$	1	**s**	$\tan 60°$	$\sqrt{3}$	**t**	$\sin 90°$	1
u	$\tan 45°$	1	**v**	$\cos 45°$	$\frac{1}{\sqrt{2}}$	**w**	$\sin 60°$	$\frac{\sqrt{3}}{2}$	**x**	$\tan 90°$	undefined
q	$\tan 60°$	$\sqrt{3}$	**r**	$\sin 30°$	$\frac{1}{2}$	**s**	$\cos 45°$	$\frac{1}{\sqrt{2}}$	**t**	$\tan 60°$	$\sqrt{3}$

5 Find the lengths of the sides labelled with letters in exact form.

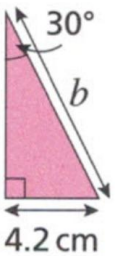
a



$\sin 30° = \frac{1}{2} = \frac{a}{15}$

$a = 7.5$ cm

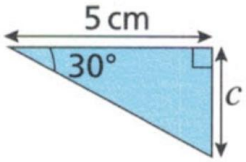
b



$\sin 30° = \frac{1}{2} = \frac{4.2}{b}$

$b = 8.4$ cm

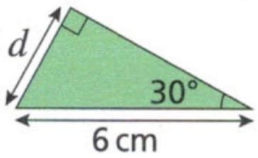
c



$\tan 30° = \frac{1}{\sqrt{3}} = \frac{c}{5}$

$c = \frac{5}{\sqrt{3}}$ cm

d

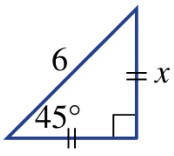


$\sin 30° = \frac{1}{2} = \frac{d}{6}$,

$d = 3$ cm

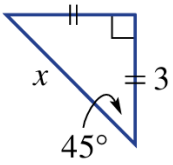
6 Find the exact value of x in each diagram. No calculators required.

a



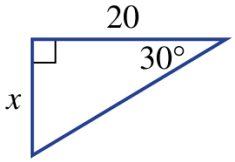
$3\sqrt{2}$

b



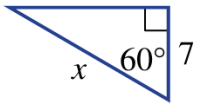
$3\sqrt{2}$

c



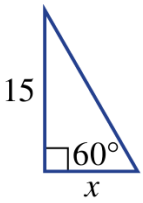
$\frac{20\sqrt{3}}{3}$

d



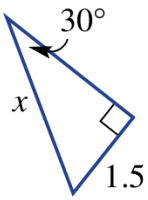
14

e



$5\sqrt{3}$

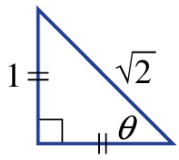
f



3

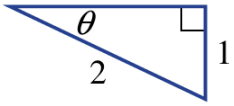
7 Find the exact value of θ in each diagram. No calculators required.

a



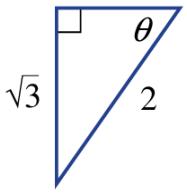
45°

b



30°

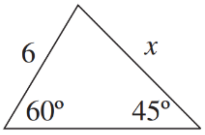
c



60°

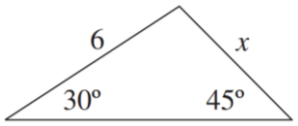
8 Find the exact value of x in each diagram. No calculators required.

a



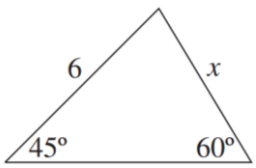
$3\sqrt{6}$

b



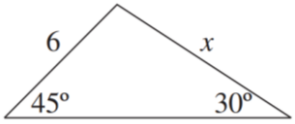
$3\sqrt{2}$

c



$2\sqrt{6}$

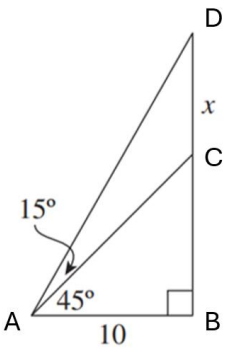
d



$6\sqrt{2}$

9

a



Show that $x = 10(\sqrt{3} - 1)$.

$\tan 60° = \frac{BD}{10}$

$BD = 10 \tan 60°$

$= 10\sqrt{3}$

$\tan 45° = \frac{BC}{10}$

$BC = 10 \tan 45°$

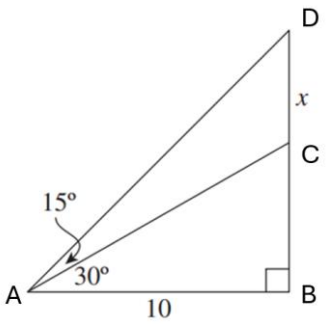
$= 10$

$x = BD - BC$

$= 10\sqrt{3} - 10$

$= 10(\sqrt{3} - 1)$

b



Show that $x = \frac{10}{3}(3 - \sqrt{3})$

$\tan 45° = \frac{BD}{10}$

$BD = 10 \tan 45°$

$= 10$

$\tan 30° = \frac{BC}{10}$

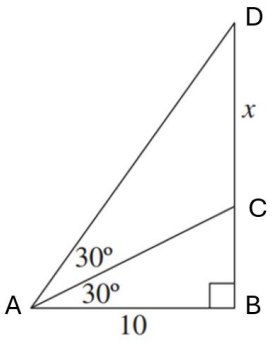
$BC = 10 \tan 30°$

$= \frac{10}{\sqrt{3}}$

$x = BD - BC$

$= \frac{10}{3}(3 - \sqrt{3})$

c



Show that $x = \frac{20}{3}\sqrt{3}$

$\tan 60° = \frac{BD}{10}$

$BD = 10 \tan 60°$

$= 10\sqrt{3}$

$\tan 30° = \frac{BC}{10}$

$BC = 10 \tan 30°$

$= \frac{10}{\sqrt{3}}$

$x = BC - BC$

$= 10\sqrt{3} - \frac{10}{\sqrt{3}}$

$= 10\sqrt{3} - \frac{10\sqrt{3}}{3}$

$= \frac{20\sqrt{3}}{3}$

10 Find the exact value of:

a $\sin 45° + \cos 30°$

$\frac{\sqrt{2} + \sqrt{3}}{2}$

b $\cos 45° + \sin 60°$

$\frac{\sqrt{2} + \sqrt{3}}{2}$

c $\sin 45° \cos 45° + \sin 30°$

1

d $\sin 60° \cos 30° - \cos 60° \sin 30°$

$\frac{1}{2}$

e $1 + \tan^2 60°$

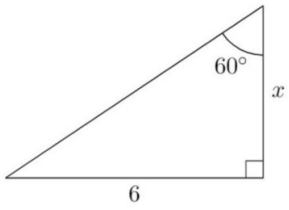
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f $\frac{1}{\sin^2 30°} - \frac{1}{\tan^2 30°}$

1

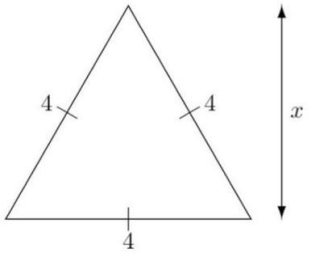
11 Find the exact value of x in these diagrams.

a



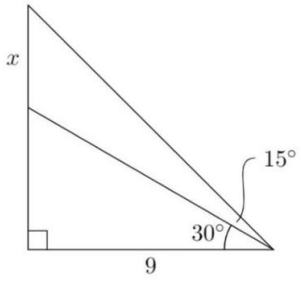
$2\sqrt{3}$

b



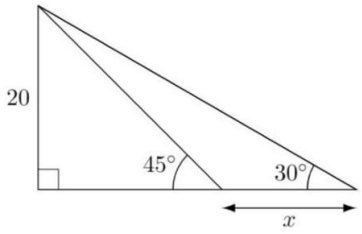
$2\sqrt{3}$

c



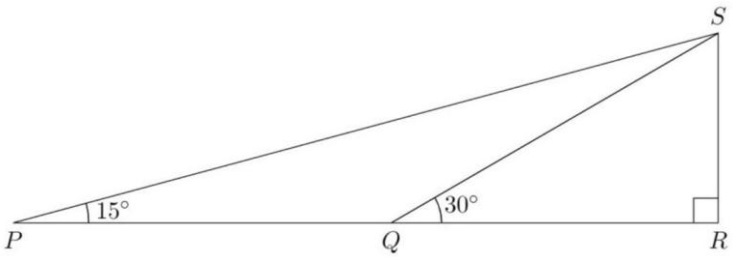
$9 - 3\sqrt{3}$

d



$20\sqrt{3} - 20$

12 Consider the diagram.



a Show that $PQ = QS$

b Given that the length of SR is 1 unit, write the lengths of QR, QS, and PQ

c Deduce that $\tan 15° = 2 - \sqrt{3}$

a) $\angle PQS = 150°$ (angles on a straight line)

$\angle PSQ = 180 - 150 - 15 = 15°$

$\therefore$ PQ = QS (sides opposite equal angles in isosceles triangles equal)

b) $\sin 30° = \frac{1}{2} = \frac{SR}{SQ} = \frac{1}{SQ}, \therefore SQ = 2, PQ = 2$

$\tan 30° = \frac{1}{\sqrt{3}} = \frac{1}{QR}, \therefore QR = \sqrt{3}$

c) $\tan 15° = \frac{SR}{PQ} = \frac{1}{2+\sqrt{3}} \frac{\times 2-\sqrt{3}}{\times 2-\sqrt{3}} = 2 - \sqrt{3}$

13 Show that:

a $2\sin 30° \cos 30° = \sin 60°$

$LHS = 2\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$
$= \frac{2\sqrt{3}}{4}$
$= \frac{\sqrt{3}}{2} = RHS$

b $\cos^2 60° - \cos^2 30° = -\frac{1}{2}$

$LHS = \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2$
$= \frac{1}{4} - \frac{3}{4}$
$= -\frac{2}{4}$
$= -\frac{1}{2} = RHS$

c $\sin 60° \cos 45° - \cos 60° \sin 45° = \frac{\sqrt{6}-\sqrt{2}}{4}$

$LHS = \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$
$= \frac{\sqrt{3}-1}{2\sqrt{2}} \frac{\times\sqrt{2}}{\times\sqrt{2}}$
$= \frac{6-\sqrt{2}}{4} = RHS$

d $\frac{\tan 60° - \tan 45°}{1 + \tan 60° \tan 45°} = 2 - \sqrt{3}$

$LHS = \frac{\sqrt{3}-1}{1+\sqrt{3}} \frac{\times 1-\sqrt{3}}{\times 1-\sqrt{3}}$
$= \frac{4-2\sqrt{3}}{2}$
$= 2 - \sqrt{3} = RHS$

e $\frac{1}{\frac{1}{\cos 30°} - \tan 30°} = \sqrt{3}$

$LHS = \frac{1}{\left(\frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}}\right)}$
$= \frac{1}{\left(\frac{1}{\sqrt{3}}\right)}$
$= \sqrt{3} = RHS$

f $\frac{1 + \tan 30°}{1 - \tan 30°} = 2 + \sqrt{3}$

$LHS = \frac{\left(1 + \frac{1}{\sqrt{3}}\right)}{\left(1 - \frac{1}{\sqrt{3}}\right)}$
$= \frac{\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)}{\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right)}$
$= \frac{\sqrt{3}+1}{\sqrt{3}-1} \frac{\times(\sqrt{3}+1)}{\times(\sqrt{3}+1)}$
$= \frac{4+2\sqrt{3}}{2}$
$= 2 + \sqrt{3} = RHS$

14 For this question, use the identity $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

a Let $t = \tan 15°$. Show that $\frac{2t}{1-t^2} = \frac{1}{\sqrt{3}}$

b Show that $t^2 + 2\sqrt{3}t - 1 = 0$

c Hence show that $\tan 15° = 2 - \sqrt{3}$

d Use a similar technique to find the exact value for $\tan 22.5°$.

$a)$ Let $\theta = 15°$, $\quad \tan 30° = \frac{2\tan 15°}{1-\tan^2 15°} = \frac{1}{\sqrt{3}}$

$\therefore \frac{2t}{1-t^2} = \frac{1}{\sqrt{3}}$

b) cross multiply

$2\sqrt{3}t = 1 - t^2$

$\therefore t^2 + 2\sqrt{3}t - 1 = 0$

c) using quadratic for t

$t = \frac{-2\sqrt{3} \pm \sqrt{12+4}}{2}$

$= -\sqrt{3} \pm 2$

$\therefore \tan 15° = -\sqrt{3} + 2$, (ignore negative as $\tan 15° > 0$)

d) $\sqrt{2} - 1$

15 For this question, use the identity $\cos 2\theta = 1 - 2\sin^2\theta$

a Let $s = \sin 15°$. Show that $s = \frac{\sqrt{2-\sqrt{3}}}{2}$

b Expand and simplify $\frac{(\sqrt{3}-1)^2}{2}$

c Hence, show that $\sin 15° = \frac{\sqrt{3}-1}{2\sqrt{2}}$

a) Let $\theta = 15°$, $\quad \cos 30° = \frac{\sqrt{3}}{2} = 1 - 2\sin^2 15°$

$\frac{\sqrt{3}}{2} = 1 - 2s^2$

$2s^2 = 1 - \frac{\sqrt{3}}{2}$

$2s^2 = \frac{2-\sqrt{3}}{2}$

$s^2 = \frac{2-\sqrt{3}}{4}$

$s = \frac{\sqrt{2-\sqrt{3}}}{2}$ (ignore negative as $\sin 15° > 0$)

b) $\frac{(\sqrt{3}-1)^2}{2} = \frac{3-2\sqrt{3}+1}{2} = 2 - \sqrt{3}$

c) $s^2 = \frac{2-\sqrt{3}}{4} = \frac{\left(\frac{(\sqrt{3}-1)^2}{2}\right)}{4} = \frac{(\sqrt{3}-1)^2}{8}$

$\therefore s = \frac{\sqrt{3}-1}{\sqrt{8}} = \frac{\sqrt{3}-1}{2\sqrt{2}}$

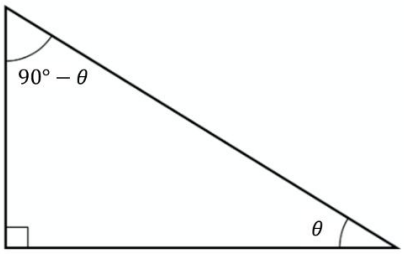
Complementary Angle Identities

ⓘ Complementary Angle Identities

For an acute angle θ, the sine of an angle equals the cosine of its complement.

$$\sin\theta = \cos(90° - \theta)$$

$$\cos\theta = \sin(90° - \theta)$$



⚠ '**co**sine' is an abbreviation for '**co**mplement of sine'

🔍 **Convert** between sine and cosine

$\sin 10° = \cos 80°$	$\sin 80° = \cos 10°$	$\cos 72° = \sin 18°$

Find the value of θ:

a	$\sin 35° = \cos\theta$	**b**	$\sin 55° = \cos\theta$	**c**	$\cos 12° = \sin\theta$
d	$\cos 15° = \sin\theta$	**e**	$\sin 89° = \cos\theta$	**f**	$\sin\theta = \cos\theta$

1 θ is acute. Find its value.

a $\sin 10° = \cos\theta$

80°

b $\sin 28° = \cos\theta$

62°

c $\cos 54° = \sin\theta$

36°

d $\sin 59° = \cos\theta$

31°

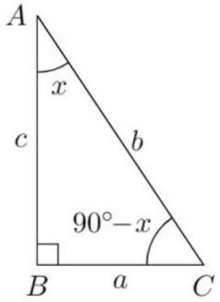
e $\cos 73° = \sin\theta$

17°

f $\cos 83° = \sin\theta$

7°

2 Use the diagram to write a ratio for the following:



a $\sin x$

$\frac{a}{b}$

b $\cos x$

$\frac{c}{b}$

c $\tan x$

$\frac{a}{c}$

d $\sin(90 - x)$

$\frac{c}{b}$

e $\cos(90 - x)$

$\frac{a}{b}$

f $\tan(90 - x)$

$\frac{c}{a}$

3 What is the relationship between $\tan x$ and $\tan(90 - x)$?

They are reciprocals of each other.

4 Simplify without a calculator.

a $\cos 22° - \sin 68°$

0

b $\frac{-\sin 57°}{\cos 53°}$

−1

c $\frac{\cos 28° + \sin 62°}{\sin 62°}$

2

d $\frac{\sin 20° + \cos 70°}{\sin 20°}$

2

e $\frac{\cos 42° + 2\sin 48°}{5\sin 48°}$

$\frac{3}{5}$

f $\sin 20° \cos 70°$

$\sin^2 20°$

5 For this question, simplify using the fact that $\tan(90 - x) = \frac{1}{\tan x}$

a $\tan 20° + \frac{1}{\tan 70°}$

$2\tan 20°$

b $\tan 30° \tan 60°$

1

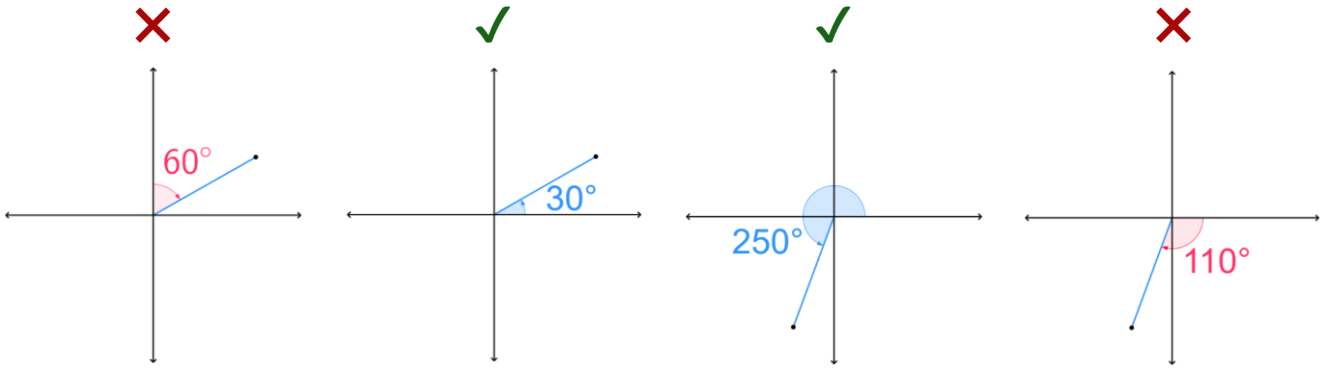
c $\tan 35° - \frac{1}{\tan 55°}$

0

Angle of Inclination

ⓘ Angle of Inclination

The angle of inclination θ is measured **anticlockwise** from **the positive x-axis** to a line.



⚠ In this booklet, unless specifically labelled otherwise, θ will refer to the angle of inclination.

ⓘ Quadrants

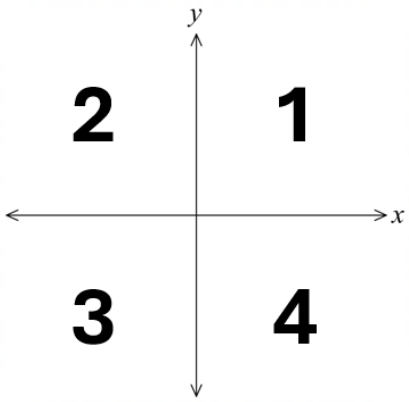
The x and y axes split the Cartesian plane into four quadrants:

Q1: $0° < \theta < 90°$

Q2: $90° < \theta < 180°$

Q3: $180° < \theta < 270°$

Q4: $270° < \theta < 360°$



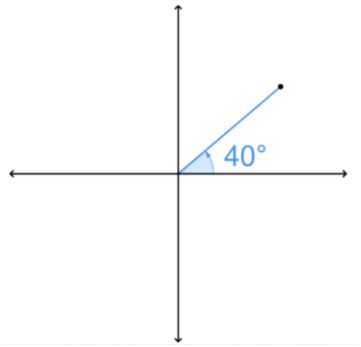
Angles of $0°, 90°, 180°, 270°$ are the **boundary angles**.

Key Ideas

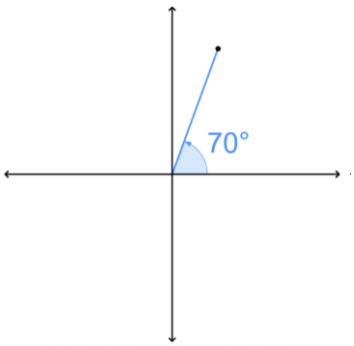
1. The angle of …… measures the rotation from the …… x-axis in an …… direction.
2. The Cartesian plane is divided into four ……

1 Sketch these angles of inclination:

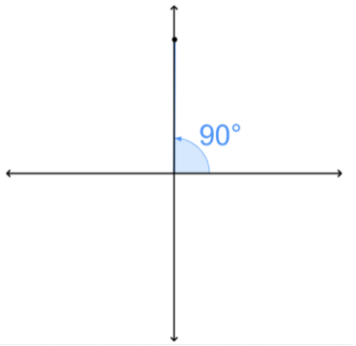
a 40°



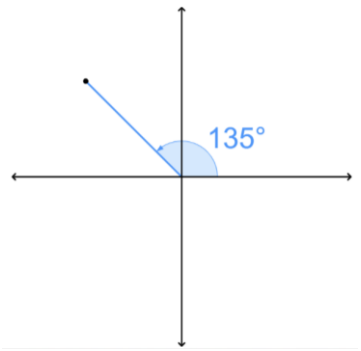
b 70°



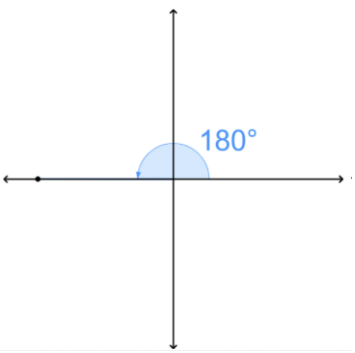
c 90°



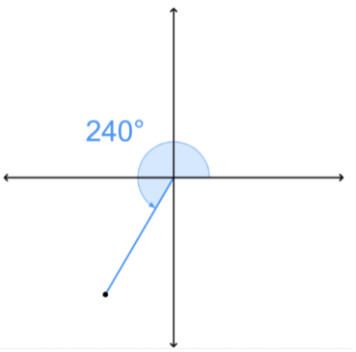
d 135°



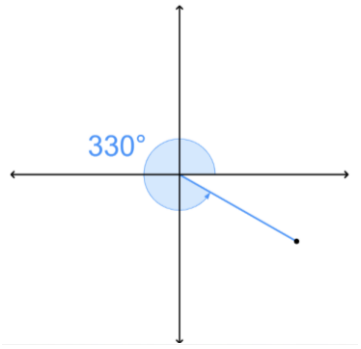
e 180°



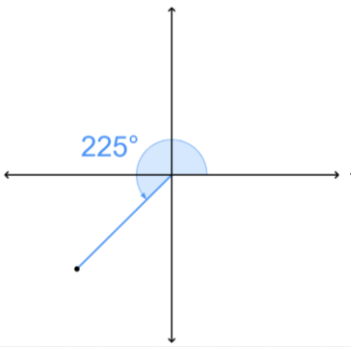
f 240°



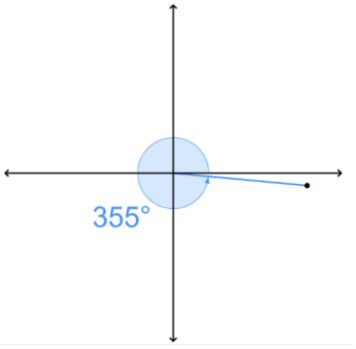
g 330°



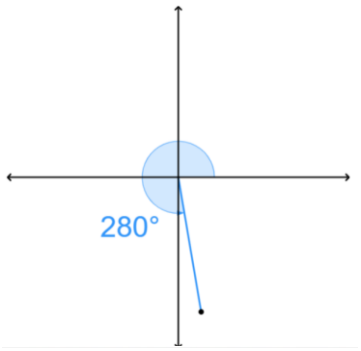
h 225°



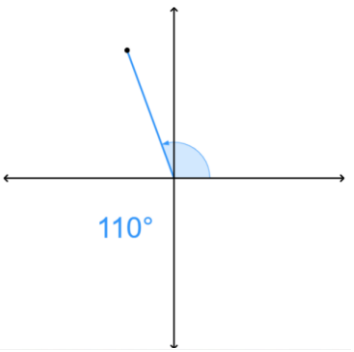
i 355°



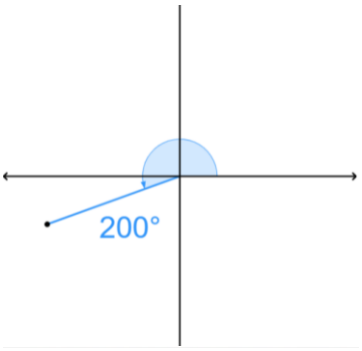
j 280°



k 110°

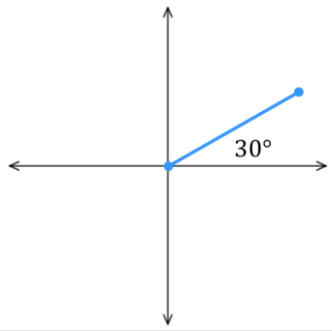


l 200°



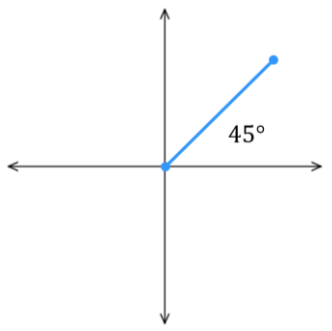
2 What is the angle of inclination between the positive x-axis and each line?

a



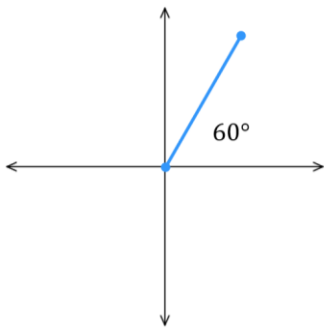
30°

b



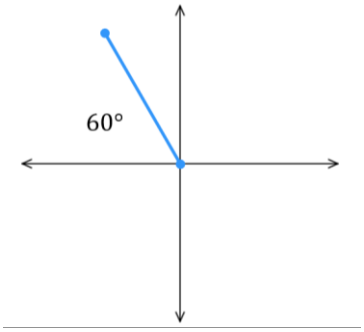
45°

c



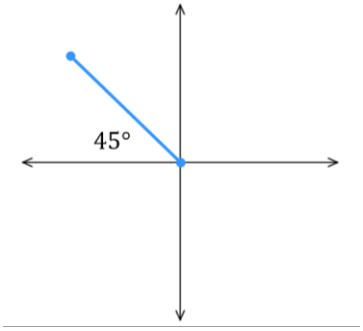
60°

d



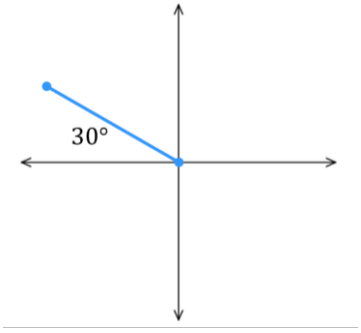
120°

e



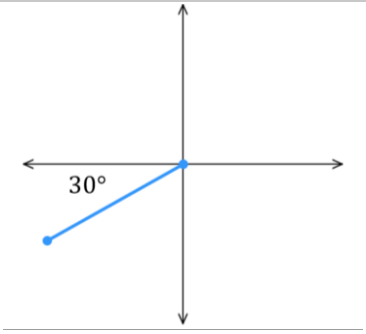
135°

f



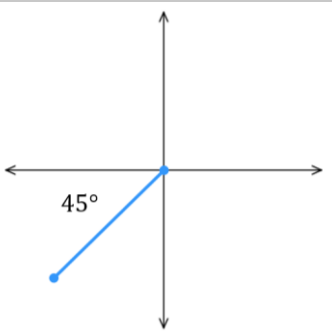
150°

g



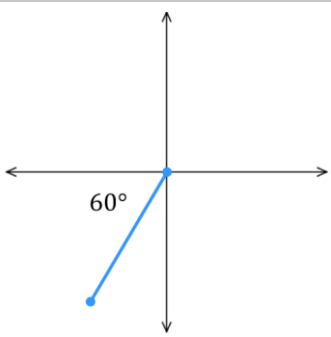
210°

h



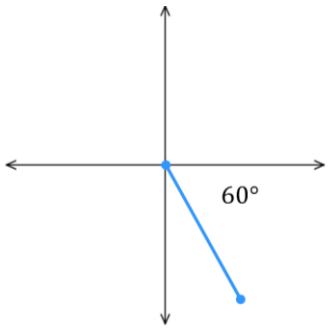
225°

i



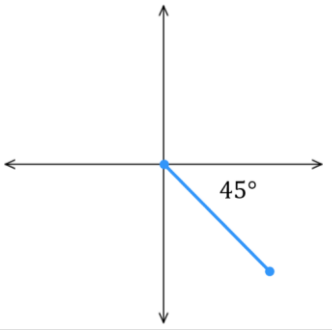
240°

j



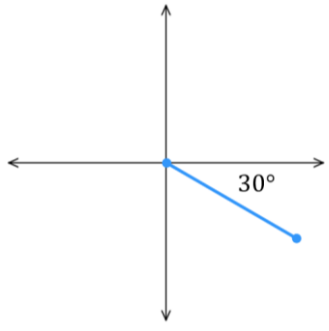
300°

k



315°

l



330°

3 Convert these true bearings to angles of inclination

a 070°

20°

b 180°

270°

c 315°

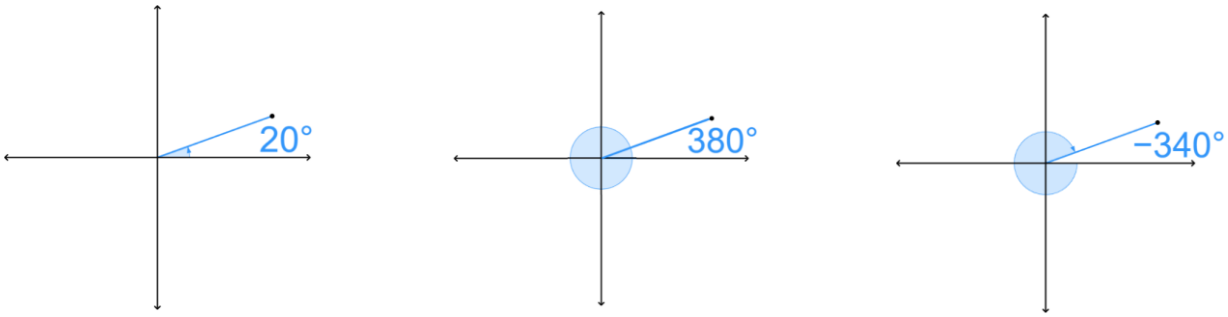
135°

Periodicity

Periodicity

To turn more than 360°, continue rotating around the circle.

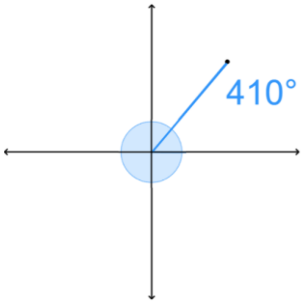
To turn a negative angle, rotate clockwise (instead of anticlockwise).



Any angle can be rewritten as an equivalent angle between 0° and 360°.

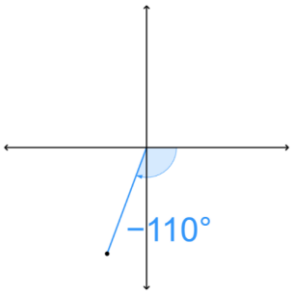
If $\theta \geq 360°$	**If $\theta < 0$**
Subtract 360° repeatedly	Add 360° repeatedly

Rewrite angles to be between 0 and 360 degrees.



$410° \equiv 410° - 360°$

$\equiv 50°$



$-110° \equiv -110° + 360°$

$\equiv 250°$

Rewrite these angles as an equivalent angle where $0 \leq \theta < 360°$

a	$400°$	**b**	$-50°$	**c**	$720°$
d	$-200°$	**e**	$900°$	**f**	$-1000°$

Key Ideas

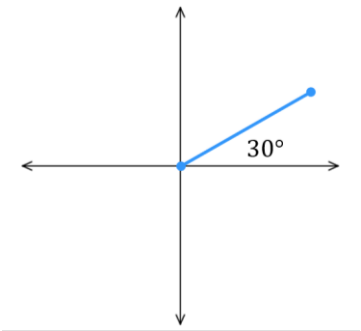
1. A full rotation around a circle is ……°, this means angles that differ by 360° represent the …… angle.
2. A negative angle represents rotation in the …… direction.
3. To rewrite an angle to the range $0 \leq \theta < 360°$:
   a. If an angle is ≥ 360°, …… 360° repeatedly.
   b. If an angle is < 0, …… 360° repeatedly.

1 Rewrite these angles of inclination as an angle $0° \leq \theta < 360°$.

a	$361°$	**b**	$-10°$	**c**	$720°$
	1°		350°		0°
d	$-500°$	**e**	$895°$	**f**	$-90°$
	220°		175°		270°
g	$1000°$	**h**	$-1000°$	**i**	$-45°$
	280°		80°		315°
j	$540°$	**k**	$-135°$	**l**	$-315°$
	180°		225°		45°
m	$660°$	**n**	$-45°$	**o**	$540°$
	300°		315°		180°
p	$-720°$	**q**	$-300°$	**r**	$1080°$
	0°		60°		0°

2

a List 6 possible angles of inclination that this ray could represent.



Many answers possible, $\theta = 30 \pm 360n$, where n is an integer.

b Is there a limit to how many angles this ray could represent?

No.

Redefining Trigonometry using the Unit Circle

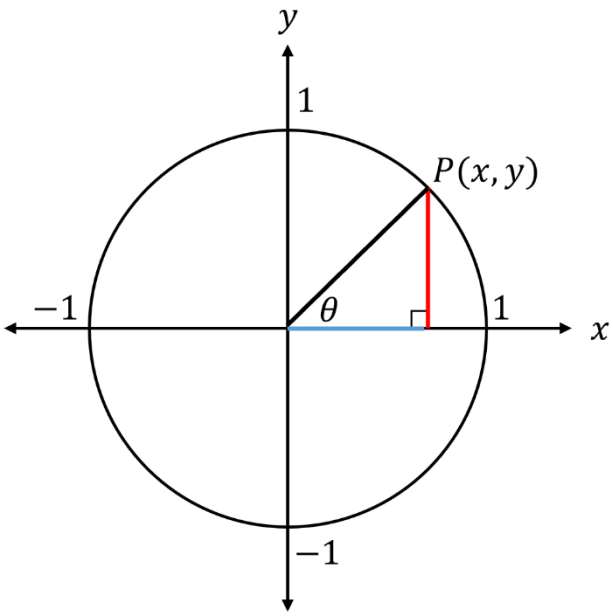
Investigation Redefining trigonometry

A **unit circle** is a circle centred at the origin (0,0) with radius 1.

On a unit circle, consider a point P at angle of inclination θ.

Draw a right triangle by dropping a perpendicular from P to the x-axis.

The hypotenuse is the radius = 1.



The horizontal leg equals the x-coordinate of P.

The vertical leg equals the y-coordinate of P.

a $\sin\theta = \frac{\text{Opp}}{\text{Hyp}} =$

$\therefore \sin\theta =$

b $\cos\theta = \frac{\text{Adj}}{\text{Hyp}} =$

$\therefore \cos\theta =$

c $\tan\theta = \frac{\text{Opp}}{\text{Adj}} =$

$\therefore \tan\theta =$

On a unit circle, consider a point $P(x, y)$ with angle of inclination θ.

- cos θ is the x-coordinate of P.
- sin θ is the y-coordinate of P.
- tan θ is the gradient of the line OP.

On a unit circle, the x-coordinate equals …… and the y-coordinate equals ……

sin θ is the …-coordinate on the unit circle, cos θ is the …-coordinate on the unit circle

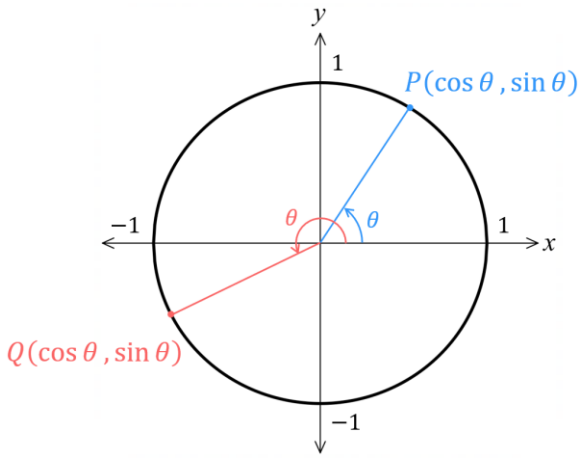
tan θ is the …… of the line.

https://www.geogebra.org/m/jzu5mddx

ⓘ Trigonometry on a Unit Circle

$P(\cos\theta, \sin\theta)$ is a point on a unit circle.

- The radius extends from the centre to P.
- θ is the angle of inclination between the positive x-axis and the radius.
- $\cos\theta$ is the x-coordinate, $\sin\theta$ is the y-coordinate.
- $\tan\theta$ is the gradient of the line OP.



$x = \cos\theta$ $\quad y = \sin\theta$ $\quad \tan\theta = \dfrac{y}{x} = \dfrac{\sin\theta}{\cos\theta}$

🔍 **Identify** Sine and cosine ratios as points on a unit circle

Show the approximate location of these points on the unit circle.

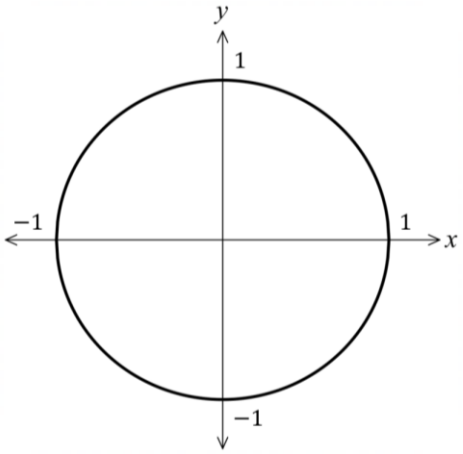
a $A(\cos 60°, \sin 60°)$

b $B(\cos 90°, \sin 90°)$

c $C(\cos 150°, \sin 150°)$

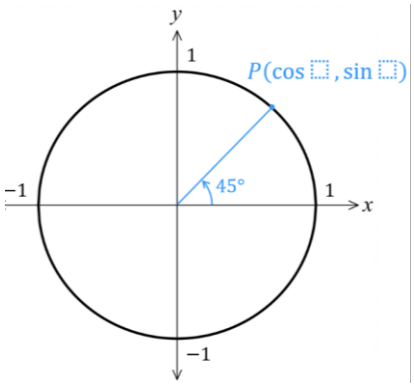
d $D(\cos 250°, \sin 250°)$

e $E(\cos 300°, \sin 300°)$

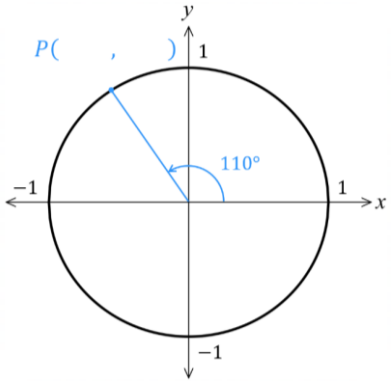


Write the coordinates of these points.

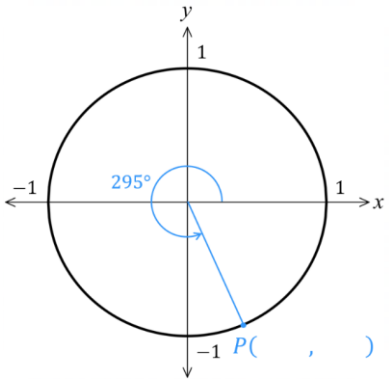
a



b



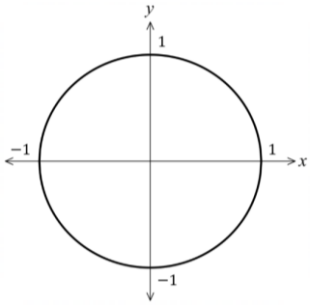
c



Interpret Sine and cosine ratios as points on a unit circle

Using the given angle of inclination, plot the point, then find the value of x and y.

a $\theta = 0°$



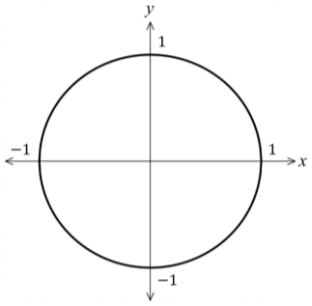
$x = \cos\theta \qquad y = \sin\theta$

$= \cos \ldots \qquad = \sin \ldots$

$= \ldots\ldots \qquad = \ldots\ldots$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

b $\theta = 90°$



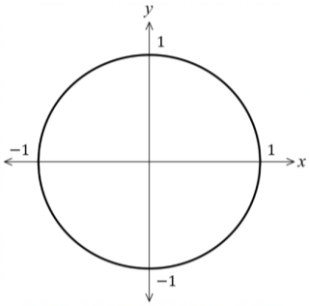
$x = \cos\theta \qquad y = \sin\theta$

$= \cos \ldots \qquad = \sin \ldots$

$= \ldots\ldots \qquad = \ldots\ldots$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

c $\theta = 180°$



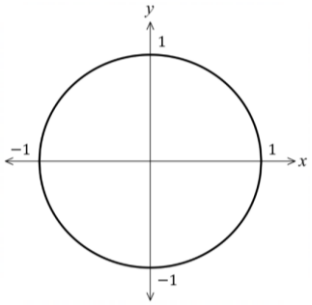
$x = \cos\theta \qquad y = \sin\theta$

$= \cos \ldots \qquad = \sin \ldots$

$= \ldots\ldots \qquad = \ldots\ldots$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

d $\theta = 270°$

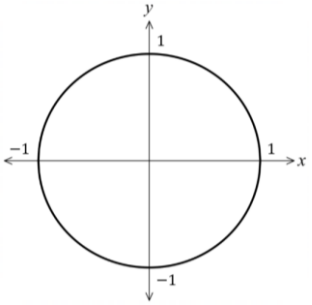


$x = \cos \ldots \qquad y = \sin \ldots$

$= \ldots\ldots \qquad = \ldots\ldots$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

e $\theta = 30°$

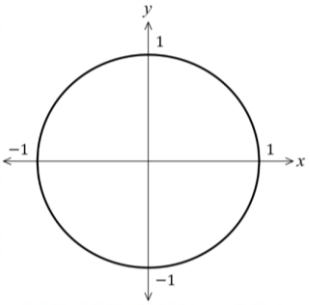


$x = \cos \ldots \qquad y = \sin \ldots$

$= \ldots\ldots \qquad = \ldots\ldots$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

f $\theta = 60°$

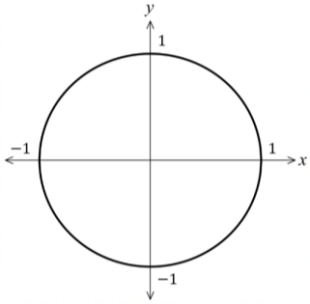


$x = \cos \ldots \qquad y = \sin \ldots$

$= \ldots\ldots \qquad = \ldots\ldots$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

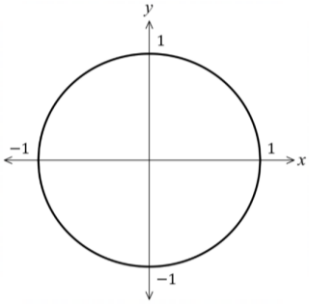
g $\theta = 45°$



$x = \qquad y =$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

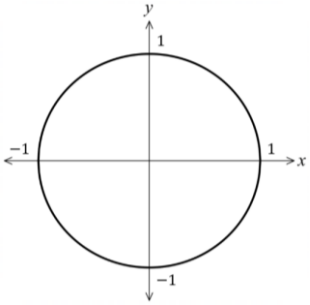
h $\theta = 390°$



$x = \qquad y =$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

i $\theta = -90°$

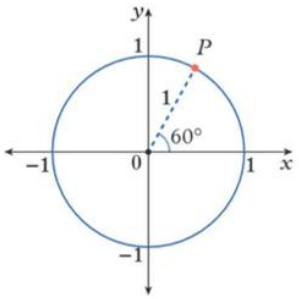


$x = \qquad y =$

$\therefore P = (\ldots\ldots, \ldots\ldots)$

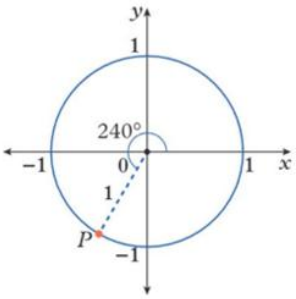
1 Find the coordinates of the point $P(\cos\theta, \sin\theta)$ on these unit circles.
Round your answer to 2 d.p. where necessary.

a



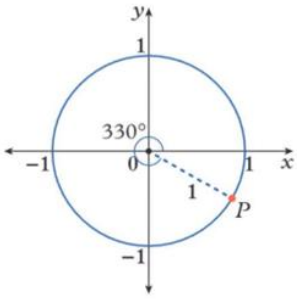
$(0.5, 0.87)$

b



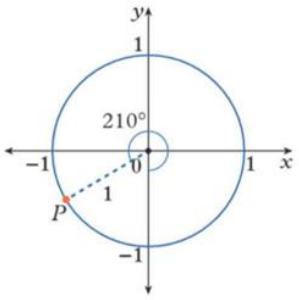
$(-0.5, -0.87)$

c



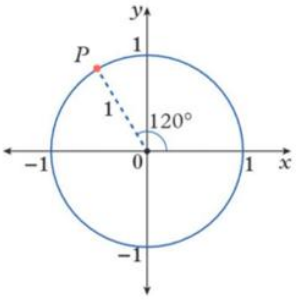
$(0.87, -0.5)$

d



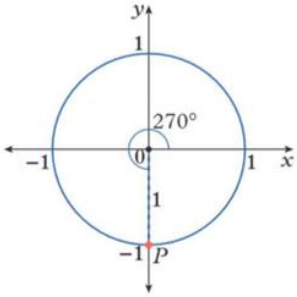
$(-0.87, -0.5)$

e



$(-0.5, -0.87)$

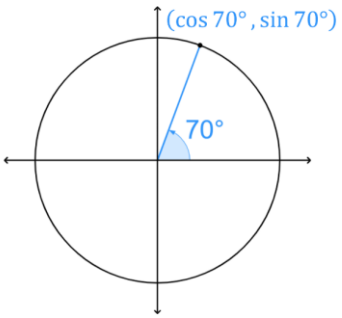
f



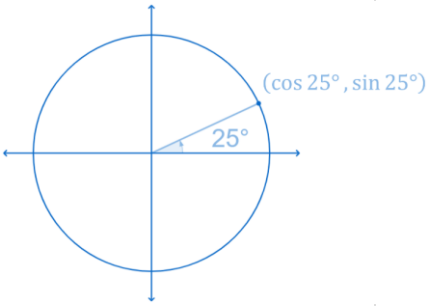
$(0, -1)$

2 Identify these points on the unit circle, showing the radius and angle of inclination.
The first question has been done for you.

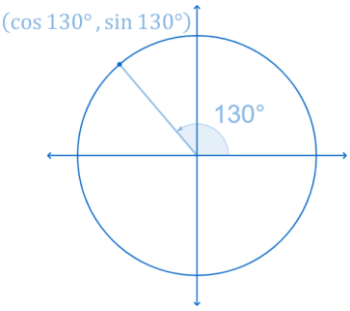
a $(\cos 70°, \sin 70°)$



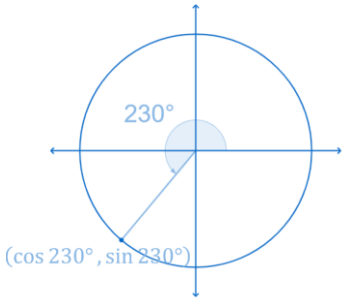
b $(\cos 25°, \sin 25°)$



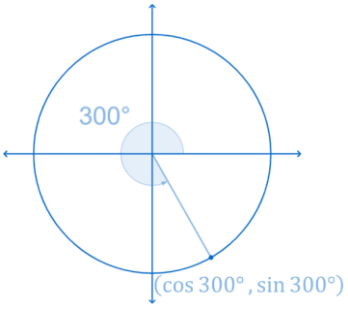
c $(\cos 130°, \sin 130°)$



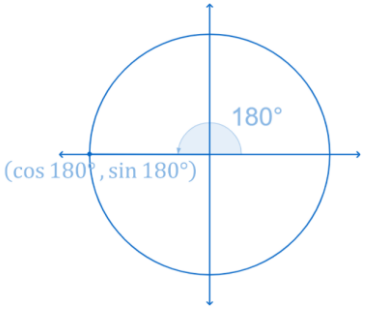
d $(\cos 230°, \sin 230°)$



e $(\cos 300°, \sin 300°)$



f $(\cos 180°, \sin 180°)$



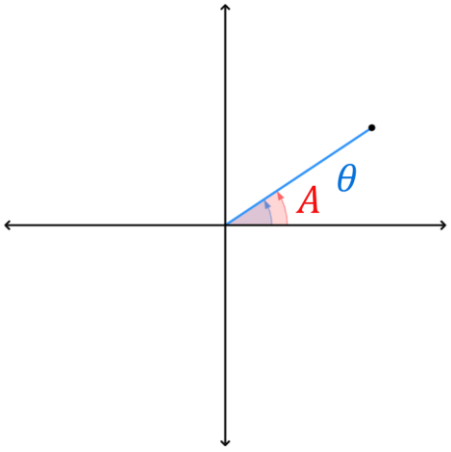
Reference Angles

ⓘ Reference Angles

The **reference angle** A is the **acute** angle between the x**-axis** and the **radius**.

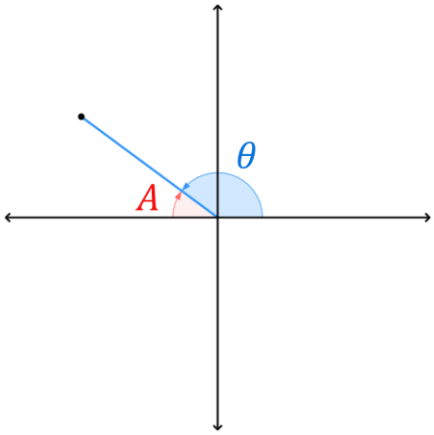
Q1

$A = \theta$



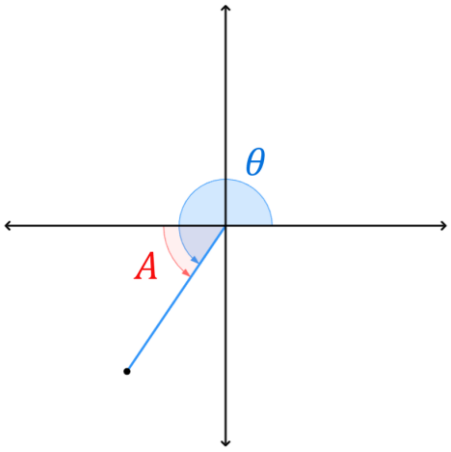
Q2

$A = 180° - \theta$



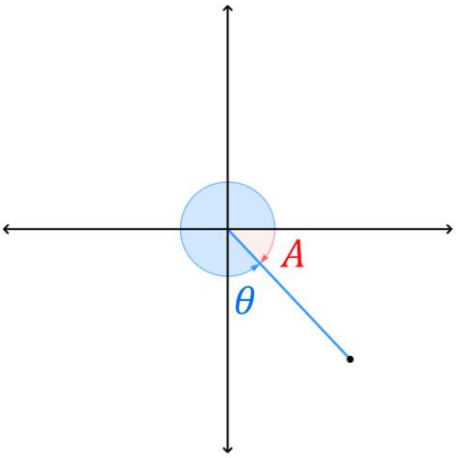
Q3

$A = \theta - 180°$



Q4

$A = 360° - \theta$



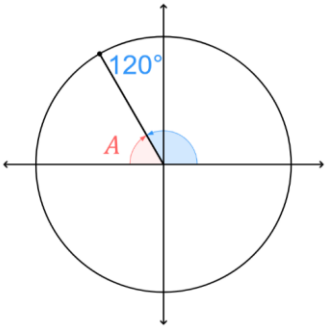
⚠ In this booklet θ refers to the angle of inclination and A the reference angle.

Key Ideas

1. The angle of inclination is how much to turn …… from the positive …-axis
2. The reference angle is the …… angle from either side of the …-axis

Rewrite Angles of inclination as a reference angle

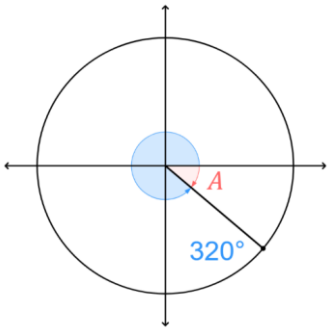
$\theta = 120°$



$A = 180° - 120°$

$= 60°$

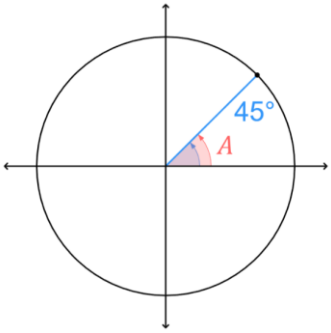
$\theta = 320°$



$A = 360° - 320°$

$= 40°$

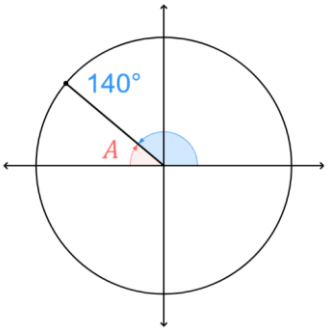
$\theta = 45°$



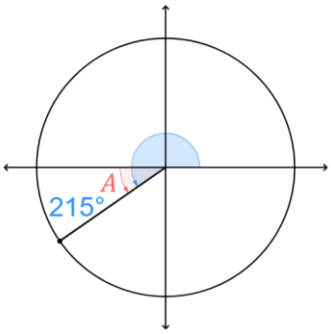
$A = 45°$

Determine the reference angle for these angles of inclination.

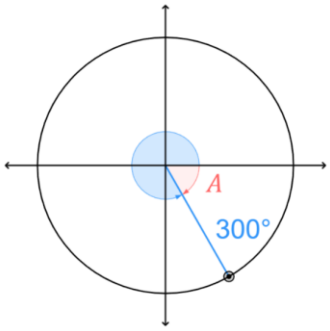
a $\theta = 140°$



b $\theta = 215°$



c $\theta = 300°$



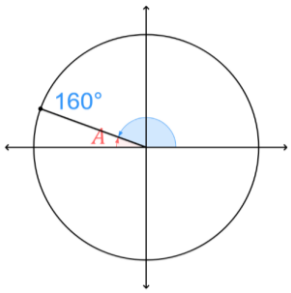
d $\theta = 150°$

e $\theta = 290°$

f $\theta = 410°$

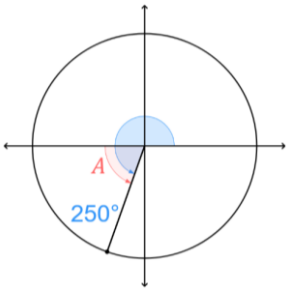
1 Find the reference angle, or related angle, of these angles of inclination. A sketch will help.

a $\theta = 160°$



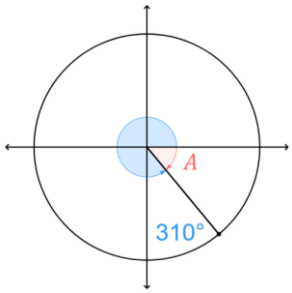
20°

b $\theta = 250°$



70°

c $\theta = 310°$



50°

d $\theta = 200°$

20°

e $\theta = 290°$

70°

f $\theta = 250°$

70°

g $\theta = 10°$

10°

h $\theta = 215°$

35°

i $\theta = 240°$

60°

j $\theta = 300°$

60°

k $\theta = 135°$

45°

l $\theta = 120°$

60°

m $\theta = 315°$

45°

n $\theta = 135°$

45°

o $\theta = 150°$

30°

p $\theta = 100°$

80°

q $\theta = 350°$

10°

r $\theta = 160°$

20°

s $\theta = 225°$

45°

t $\theta = 210°$

30°

u $\theta = 300°$

60°

v $\theta = 420°$

60°

w $\theta = -300°$

60°

x $\theta = -135°$

45°

y $\theta = 495°$

45°

z $\theta = -30°$

30°

aa $\theta = 675°$

45°

Trigonometric Ratios of Obtuse Angles

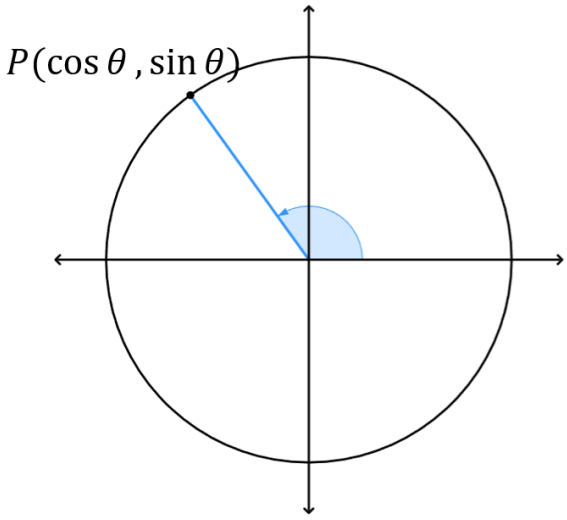
Investigation Sign of trigonometric ratios in Q2

In Q2, the point P has a negative x-coordinate and a positive y-coordinate.

Since cos θ = x-coordinate, and x is negative in Q2, cos θ is negative.

Since sin θ = y-coordinate, and y is positive in Q2, sin θ is positive.

Since $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\text{positive}}{\text{negative}}$, tan θ is negative.



In Q2, sin is ……, cos is ……, tan is ……

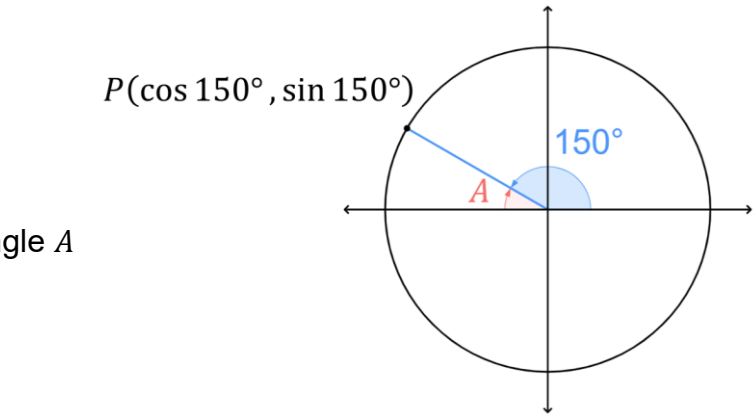
The only positive trig ratio in Q2 is ……

Investigation Trigonometric ratios for obtuse angles

Consider θ = 150°.

The reference angle is $A =$ ……

Construct a right-angled triangle with angle A in the diagram to the right.



What do you notice about the value of $\sin 30°$ and $\sin 150°$?

$\sin 150°$ *is …… to the value of* $\sin 30°$.

What do you notice about the value of $\cos 30°$ and $\cos 150°$?

$\cos 150°$ *is …… to the value of* $\cos 30°$, *except it is ……*

Therefore, we can write:

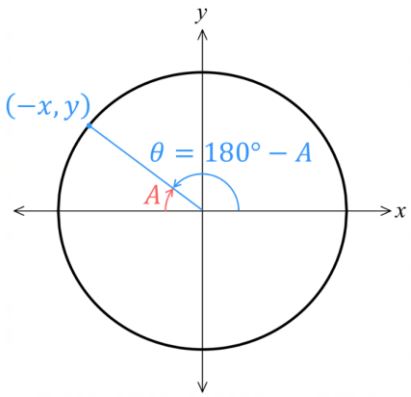
$$\sin 150° = \sin 30°$$

$$\cos 150° = -\cos 30°$$

$$\tan 150° = \frac{\sin 150°}{\cos 150°} =$$

Trigonometric Ratios for Obtuse Angles

We express trigonometric ratios for **obtuse angles θ** according to its acute reference angle A:



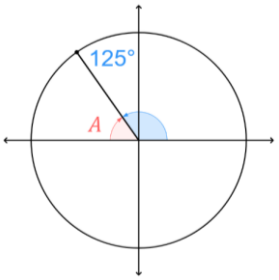
$$\sin\theta = \sin A$$
$$\cos\theta = -\cos A$$
$$\tan\theta = -\tan A$$

To rewrite trigonometric ratios with the reference angle:

1. Find the reference angle (in Q2, $A = 180° - \theta$).
2. Determine the sign (in Q2, sin is positive, cos and tan are negative).
3. Rewrite trig ratio with acute reference angle and the correct sign.

Example Rewrite trigonometric ratios with the acute reference angle (2nd quadrant).

Write $\sin 125°$ using its reference angle.



1. $A = 55°$
2. sin is positive in Q2
3. $\sin 125° = \sin 55°$

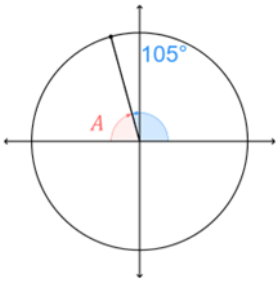
How did we know $A = 55°$?

Angles on a ……-line sum to ……

Why is $\sin 125°$ positive?

The …-coordinate in the …… quadrant of a unit circle is positive.

Write $\cos 105°$ using its reference angle.



1. $A = 75°$
2. cos is negative in Q2
3. $\cos 105° = -\cos 75°$

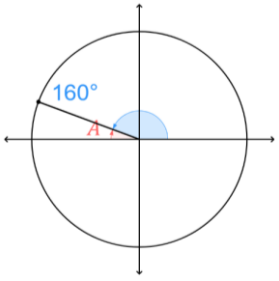
Why is $\cos 105°$ negative?

The …-coordinate in the …… quadrant of a unit circle is negative.

Guided Practice

Rewrite these trigonometric ratios in terms of its acute reference angle.

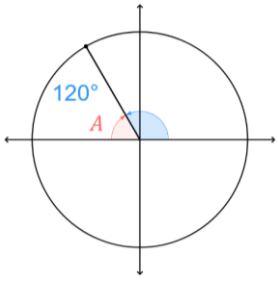
a $\sin 160°$



1. $A =$
2. sign:
3. $\sin 160° =$

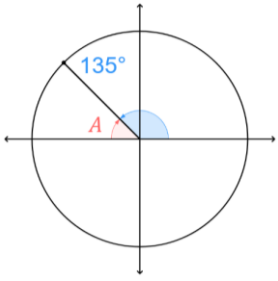
$\sin 20°$

b $\cos 120°$



$-\cos 60°$

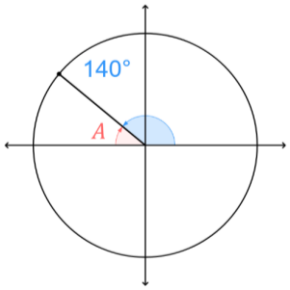
c $\tan 135°$



$-\tan 45°$

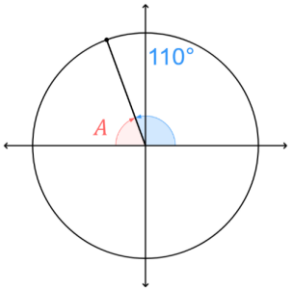
1 Rewrite these trigonometric ratios in terms of its acute reference angle.

a $\cos 140°$



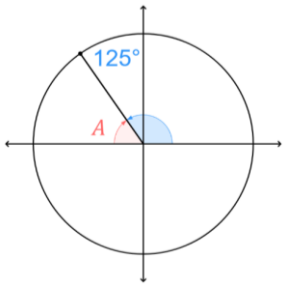
$-\cos 40°$

b $\sin 110°$



$\sin 70°$

c $\cos 125°$



$-\cos 55°$

d $\cos 175°$

$-\cos 5°$

e $\tan 170°$

$-\tan 10°$

f $\tan 95°$

$-\tan 85°$

g $\sin 175°$

$\sin 5°$

h $\sin 180°$

$\sin 0°$

i $\cos 155°$

$-\cos 25°$

j $\tan 120°$

$-\tan 60°$

k $\cos 150°$

$-\cos 30°$

l $\sin 160°$

$\sin 20°$

m $\sin 140°$

$\sin 40°$

n $\cos 125°$

$-\cos 55°$

o $\tan 123°$

$-\tan 57°$

p $\tan 105°$

$-\tan 75°$

q $\cos 155°$

$-\cos 25°$

r $\tan 100°$

$-\tan 80°$

s $\cos 148°$

$-\cos 32°$

t $\sin 167°$

$\sin 13°$

u $\tan 129°$

$-\tan 51°$

2 Find the exact values of these trigonometric ratios by first rewriting them with the acute reference angle.

a $\sin 150°$

$= \sin 30°$

$= \ldots\ldots\ldots$

$\frac{1}{2}$

b $\cos 150°$

$= -\cos 30°$

$= \ldots\ldots\ldots$

$-\frac{\sqrt{3}}{2}$

c $\tan 120°$

$-\sqrt{3}$

d $\cos 135°$

$-\frac{1}{\sqrt{2}}$

e $\sin 120°$

$\frac{\sqrt{3}}{2}$

f $\cos 120°$

$-\frac{1}{2}$

g $\tan 150°$

$-\frac{1}{\sqrt{3}}$

h $\sin 135°$

$\frac{1}{\sqrt{2}}$

i $\tan 135°$

-1

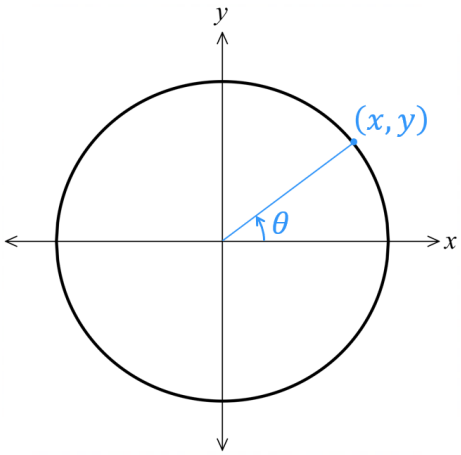
Trigonometric Ratios of Angles of Any Magnitude

ⓘ Angles of Any Magnitude

You can express trigonometric ratios of any angle using the acute reference angle A.

Q1

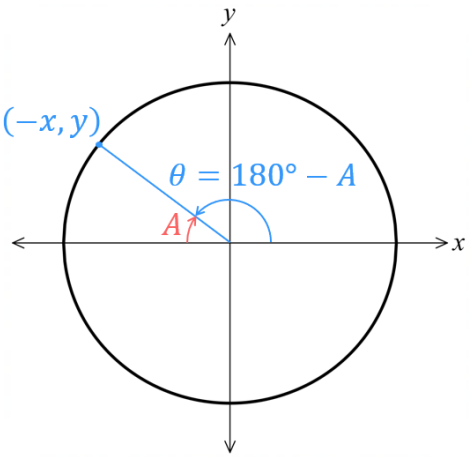
$A = \theta$



$\sin\theta = \sin A$
$\cos\theta = \cos A$
$\tan\theta = \tan A$

Q2

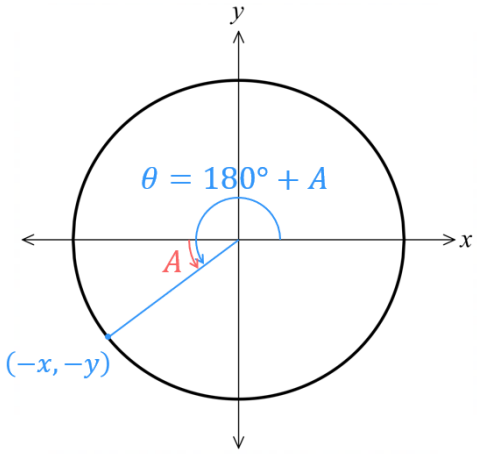
$A = 180° - \theta$
$\theta = 180° - A$



$\sin(180° - A) = \sin A$
$\cos(180° - A) = -\cos A$
$\tan(180° - A) = -\tan A$

Q3

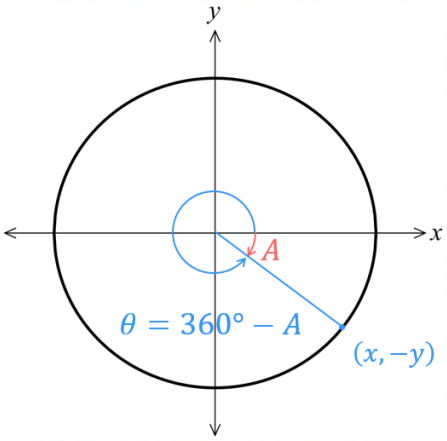
$A = \theta - 180°$
$\theta = 180° + A$



$\sin(180° + A) = -\sin A$
$\cos(180° + A) = -\cos A$
$\tan(180° + A) = \tan A$

Q4

$A = 360° - \theta$
$\theta = 360° - A$



$\sin(360° - A) = -\sin A$
$\cos(360° - A) = \cos A$
$\tan(360° - A) = -\tan A$

ⓘ ASTC

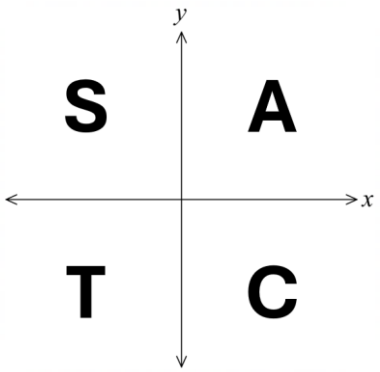
All the identities on the previous page can be memorised using ASTC:

All positive in 1st quadrant.

Sin positive in 2nd quadrant; others are negative.

Tan positive in 3rd quadrant; others are negative.

Cos positive in 4th quadrant; others are negative.



Q1	Q2	Q3	Q4
$\theta = 180 - A$ $A = 180 - \theta$	$\theta = 180 + A$ $A = \theta - 180$	$\theta = 180 + A$ $A = \theta - 180$	$\theta = 360 - A$ $A = 360 - \theta$
All positive	**S**in positive, others negative	**T**an positive, others negative	**C**os positive, others negative

🔍 **Identify** sign of trigonometric ratios of any angle

Determining the sign:

1. Identify the quadrant of θ.
2. Use ASTC to determine if the ratio is positive or negative.

$\sin 200°$	$\tan 200°$	$\cos 320°$
$200°$ is in Q3 $\therefore \sin 200°$ is negative	$200°$ is in Q3 $\therefore \tan 200°$ is positive	$320°$ is in Q4 $\therefore \cos 320°$ is positive

Determine the sign of these trigonometric ratios

a $\sin 320°$	**b** $\cos 250°$	**c** $\tan 140°$
d $\cos 60°$	**e** $\cos(-50°)$	**f** $\sin(700°)$

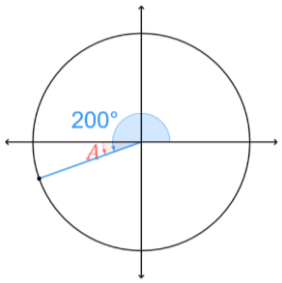
Rewriting Trigonometric Ratios with Acute Reference Angle

To rewrite trigonometric ratios with the reference angle:

1. Find the reference angle.
2. Determine the sign (using ASTC).
3. Rewrite trig ratio with acute reference angle and the correct sign.

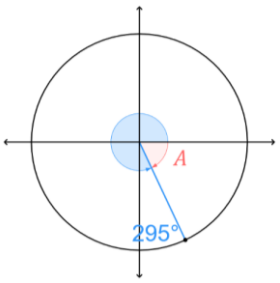
Example Rewrite trigonometric ratios with the acute reference angle (All quadrants).

Write $\sin 200°$ using its reference angle.



1. $A = 20°$
2. sin is negative in Q3
3. $\sin 200° = -\sin 20°$

Write $\cos 295°$ using its reference angle.

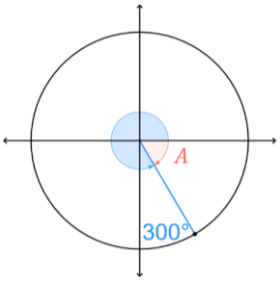


1. $A = 75°$
2. cos is positive in Q4
3. $\cos 295° = \cos 75°$

Guided Practice

Rewrite these trigonometric ratios in terms of its acute reference angle.

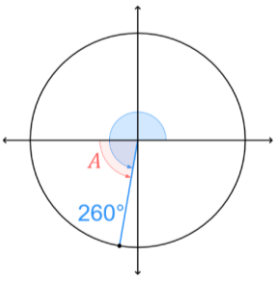
a $\sin 300°$



1. $A =$
2. sign:
3. $\sin 300° =$

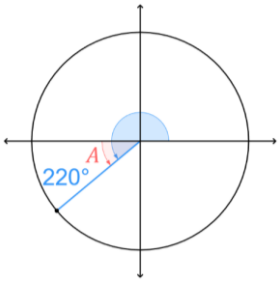
$-\sin 60°$

b $\cos 260°$



$-\cos 80°$

c $\tan 220°$



$\tan 40°$

Key Ideas

1. In quadrant 1, …… ratios are positive.
2. In quadrant 2, only …… is positive.
3. In quadrant 3, only …… is positive.
4. In quadrant 4, only …… is positive.
5. We can rewrite a trigonometric ratio of any angle using its acute …… angle by considering the quadrant and sign.

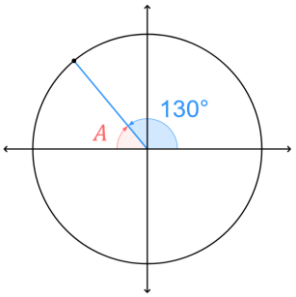
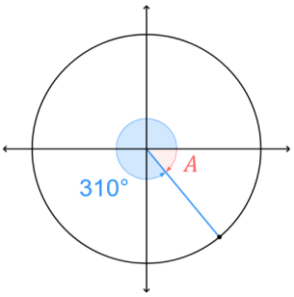
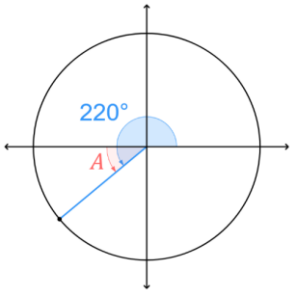
1 In which quadrants are these statements true?

a	sin θ is positive quadrants 1 and 2	**b**	tan θ is negative 2 and 4	**c**	cos θ is negative 2 and 3
d	cos θ is positive 1 and 4	**e**	tan θ is positive 1 and 3	**f**	sin θ is negative 3 and 4

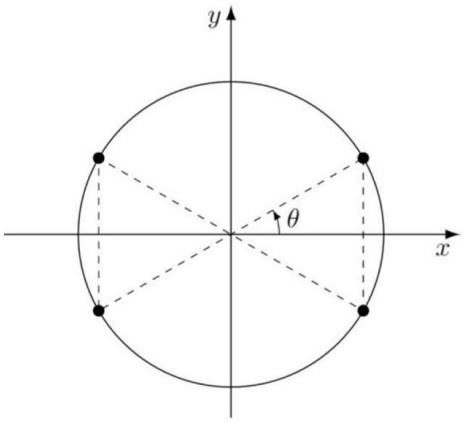
2 Use ASTC to determine whether these trigonometric ratios are positive or negative.

a	$\sin 20°$ p	**b**	$\cos 50°$ p	**c**	$\cos 100°$ n	**d**	$\tan 140°$ n
e	$\tan 250°$ p	**f**	$\sin 310°$ n	**g**	$\sin 200°$ n	**h**	$\cos 280°$ p
i	$\sin 340°$ n	**j**	$\cos 350°$ p	**k**	$\tan 290°$ n	**l**	$\cos 190°$ n
m	$\tan 170°$ n	**n**	$\sin 110°$ p	**o**	$\tan 80°$ p	**p**	$\cos 170°$ n

3 Rewrite these trigonometric ratios in terms of its acute reference angle.

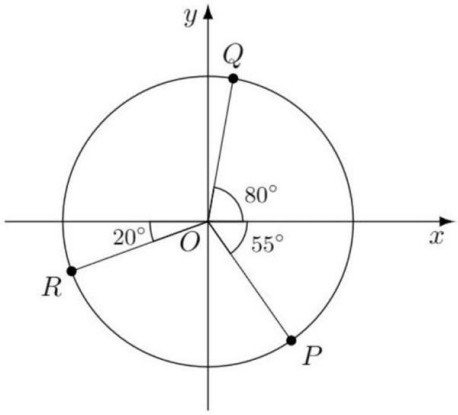
a	$\tan 130°$  $-\tan 50°$	**b**	$\cos 310°$  $\cos 50°$	**c**	$\sin 220°$  $-\sin 40°$
d	$\tan 210°$ $\tan 30°$	**e**	$\tan 315°$ $-\tan 45°$	**f**	$\cos 240°$ $-\cos 60°$
g	$\tan 260°$ $\tan 80°$	**h**	$\cos 170°$ $-\cos 10°$	**i**	$\sin 320°$ $-\sin 40°$
j	$\cos 185°$ $-\cos 5°$	**k**	$\sin 125°$ $\sin 55°$	**l**	$\tan 325°$ $-\tan 35°$
m	$\sin 85°$ $\sin 85°$	**n**	$\cos 95°$ $-\cos 85°$	**o**	$\tan 205°$ $\tan 25°$

4 Use the diagram to write these trigonometric ratios in terms of the acute angle, θ.



a	$\sin(180-\theta)$	**b**	$\cos(180-\theta)$	**c**	$\tan(180-\theta)$
	sin θ		-cos θ		-tan θ
d	$\sin(180+\theta)$	**e**	$\cos(180+\theta)$	**f**	$\tan(180+\theta)$
	-sin θ		-cos θ		tan θ
g	$\sin(360-\theta)$	**h**	$\cos(360-\theta)$	**i**	$\tan(360-\theta)$
	-sin θ		cos θ		-tan θ
j	$\sin(-\theta)$	**k**	$\cos(-\theta)$	**l**	$\tan(-\theta)$
	-sin θ		cos θ		-tan θ

5 The points P, Q, R are on the unit circle. O is the origin.
Find the coordinates of P, Q, R
Leave your answer in terms of $\cos\theta$ and $\sin\theta$, where θ is the acute reference angle.



P (cos 55°, −sin 55°)
Q (cos 80°, sin 80°)
R (−cos 20°, −sin 20°)

6 Simplify:

a	$\frac{\sin(180°+\beta)}{\cos(360°-\beta)}$	**b**	$\frac{\cos(180°-\beta)}{\sin(90°-\beta)}$	**c**	$\sin(180°+\beta)\cos(90°-\beta)$
	$-\tan\beta$		-1		$-\sin^2\beta$

7 Given that sin 25° ≈ 0.42 and cos 25° ≈ 0.91, write down approximate values, without using a calculator, for:

a	$\sin 155°$	**b**	$\cos 205°$	**c**	$\cos 335°$
	$= \sin(180° - 25°)$		$= \cos(180° + 25°)$		
	$= \sin(\quad °)$		$= -\cos(\quad °)$		
	$=$		$=$		
	0.42		−0.91		0.91
d	$\sin 335°$	**e**	$\sin 205° - \cos 155°$	**f**	$\cos 385° - \sin 515°$
	−0.42		0.49		0.49

Exact Values of Trigonometric Ratios (All Angles)

Review

☐ Recall exact values of trigonometric ratios

a	$\sin 30°$	**b**	$\cos 90°$	**c**	$\tan 60°$
d	$\cos 60°$	**e**	$\sin 45°$	**f**	$\tan 45°$
g	$\tan 30°$	**h**	$\sin 60°$	**i**	$\cos 0°$

☐ Rewrite trigonometric ratios with the acute reference angle

a	$\tan 210°$	**b**	$\cos 120°$	**c**	$\sin 300°$
d	$\tan 315°$	**e**	$\sin 225°$	**f**	$\cos 150°$

ⓘ Exact Values of any Trigonometric Ratio (Multiples of 0° 30° 45° 60° 90°)

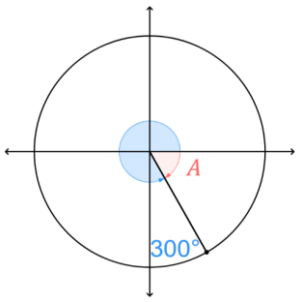
1. Rewrite trig ratio with acute reference angle and the correct sign.
2. Write with exact value using the table below:

	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

⚠ A sketch of the angle is very useful.

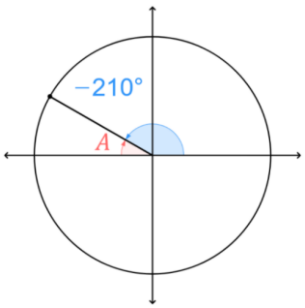
Example Find the exact value of a trigonometric function (all quadrants)

Find the exact value of $\tan 300°$



1. $\tan 300° = -\tan 60°$
2. $= -\sqrt{3}$

Find the exact value of $\sin(-210)°$

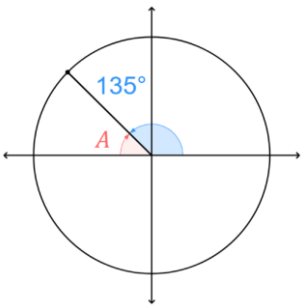


1. $\sin(-210)° = \sin 150°$
   $= \sin 30°$
2. $= \frac{1}{2}$

Guided Practice

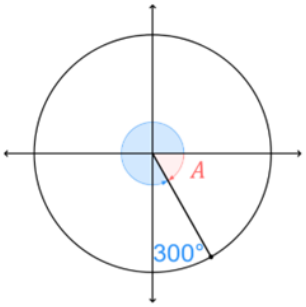
Find the exact value of:

a $\sin 135°$



$\frac{1}{\sqrt{2}}$

b $\cos 300°$



0.5

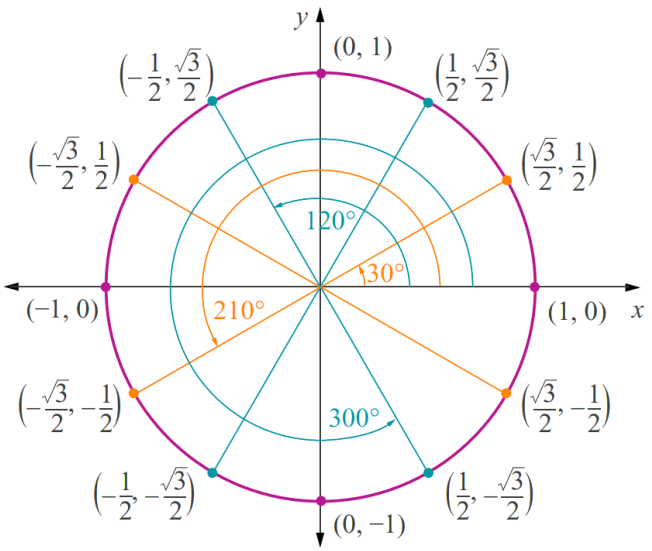
c $\sin 270°$

-1

d $\tan(-60)°$

$-\sqrt{3}$

1 The unit circle shown has exact values.
Use the unit circle to find the exact value of these trigonometric functions.

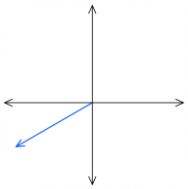
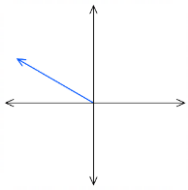
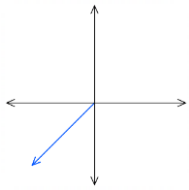
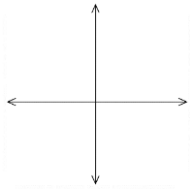
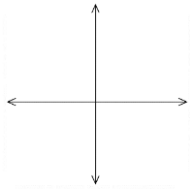
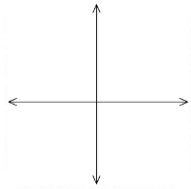
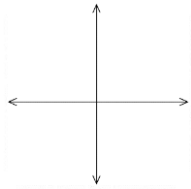
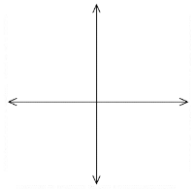
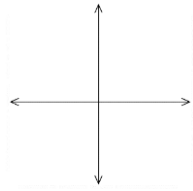


a	$\sin 0°$	**b**	$\cos 0°$	**c**	$\tan 0°$
	0		1		0
d	$\sin 120°$	**e**	$\cos 120°$	**f**	$\tan 120°$
	$\frac{\sqrt{3}}{2}$		$-\frac{1}{2}$		$-\sqrt{3}$
g	$\sin 300°$	**h**	$\cos 300°$	**i**	$\tan 300°$
	$-\frac{\sqrt{3}}{2}$		$\frac{1}{2}$		$-\sqrt{3}$
j	$\cos 210°$	**k**	$\sin 240°$	**l**	$\tan 150°$
	$-\frac{\sqrt{3}}{2}$		$-\frac{\sqrt{3}}{2}$		$-\frac{1}{\sqrt{3}}$
m	$\sin 60°$	**n**	$\cos 30°$	**o**	$\tan 240°$
	$\frac{\sqrt{3}}{2}$		$\frac{\sqrt{3}}{2}$		$\sqrt{3}$
p	$\cos 150°$	**q**	$\cos 90°$	**r**	$\sin 180°$
	$-\frac{\sqrt{3}}{2}$		0		0
s	$\sin 210°$	**t**	$\tan 330°$	**u**	$\tan 270°$
	$-\frac{1}{2}$		$-\frac{1}{\sqrt{3}}$		undefined

2 Use the unit circle to find all the angles $0° \leq \theta < 360°$ that have these values:

a	$\cos\theta = \frac{\sqrt{3}}{2}$ $\theta = 30°$ and ……°	**b**	$\sin\theta = \frac{1}{2}$	**c**	$\sin\theta = -\frac{1}{2}$
	30° and 330°		30° and 150°		210° and 330°
d	$\sin\theta = \frac{\sqrt{3}}{2}$	**e**	$\sin\theta = -\frac{\sqrt{3}}{2}$	**f**	$\cos\theta = \frac{1}{2}$
	60° and 120°		240° and 300°		60° and 300°
g	$\cos\theta = 0$	**h**	$\sin\theta = -1$	**i**	$\cos\theta = 1$
	90° and 270°		270°		0°

3 Complete the table.

Expression	Diagram	Reference Angle	Positive or Negative	Equivalent Expression	Exact Value
a $\sin 210°$		$30°$	$-$	$-\sin 30°$	$-\frac{1}{2}$
b $\sin 150°$		$30°$	$+$	$\sin 30°$	$\frac{1}{2}$
c $\sin 225°$		$45°$	$-$	$-\sin 45°$	$-\frac{1}{\sqrt{2}}$
d $\cos 300°$		$60°$	$+$	$\cos 60°$	$\frac{1}{2}$
e $\cos 150°$		$30°$	$-$	$-\cos 30°$	$-\frac{\sqrt{3}}{2}$
f $\cos 300°$		$60°$	$+$	$\cos 60°$	$\frac{1}{2}$
g $\tan 300°$		$60°$	$-$	$-\tan 60°$	$-\sqrt{3}$
h $\tan 135°$		$45°$	$-$	$-\tan 45°$	-1
i $\tan 240°$		$60°$	$+$	$\tan 60°$	$\sqrt{3}$

4 Find the exact value of these trigonometric ratios.

a	$\sin 60°$	$\frac{\sqrt{3}}{2}$	**b**	$\sin 120°$	$\frac{\sqrt{3}}{2}$	**c**	$\sin 240°$	$-\frac{\sqrt{3}}{2}$	**d**	$\sin 300°$	$-\frac{\sqrt{3}}{2}$
e	$\cos 45°$	$\frac{1}{\sqrt{2}}$	**f**	$\cos 135°$	$-\frac{1}{\sqrt{2}}$	**g**	$\cos 225°$	$-\frac{1}{\sqrt{2}}$	**h**	$\cos 315°$	$\frac{1}{\sqrt{2}}$
i	$\tan 30°$	$\frac{1}{\sqrt{3}}$	**j**	$\tan 150°$	$-\frac{1}{\sqrt{3}}$	**k**	$\tan 210°$	$\frac{1}{\sqrt{3}}$	**l**	$\tan 330°$	$-\frac{1}{\sqrt{3}}$

5 Find the exact value of these trigonometric ratios.

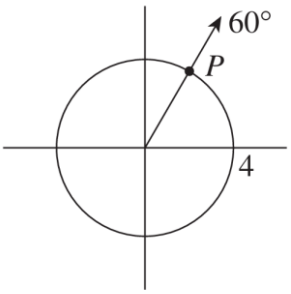
a	$\cos 120°$	$-\frac{1}{2}$	**b**	$\tan 225°$	1	**c**	$\sin 300°$	$-\frac{\sqrt{3}}{2}$	**d**	$\sin 135°$	$\frac{1}{\sqrt{2}}$
e	$\tan 240°$	$\sqrt{3}$	**f**	$\cos 210°$	$-\frac{\sqrt{3}}{2}$	**g**	$\tan 315°$	-1	**h**	$\cos 300°$	$\frac{1}{2}$
i	$\sin 225°$	$-\frac{1}{\sqrt{2}}$	**j**	$\cos 150°$	$-\frac{\sqrt{3}}{2}$	**k**	$\sin 210°$	$-\frac{1}{2}$	**l**	$\tan 300°$	$-\sqrt{3}$

6 Find the exact value of these trigonometric ratios.

a	$\cos(-60°)$	$\frac{1}{2}$	**b**	$\sin(-120°)$	$-\frac{\sqrt{3}}{2}$	**c**	$\tan(-120°)$	$\sqrt{3}$	**d**	$\sin(-315°)$	$\frac{1}{\sqrt{2}}$
e	$\tan(-210°)$	$-\frac{1}{\sqrt{3}}$	**f**	$\cos(-225°)$	$-\frac{1}{\sqrt{2}}$	**g**	$\tan 420°$	$\sqrt{3}$	**h**	$\cos 510°$	$-\frac{\sqrt{3}}{2}$
i	$\sin 495°$	$\frac{1}{\sqrt{2}}$	**j**	$\sin 690°$	$-\frac{1}{2}$	**k**	$\cos 600°$	$-\frac{1}{2}$	**l**	$\tan 585°$	1

8 When the radius of the circle is not 1, the coordinates of a point on the circumference is given by $(r\cos\theta, r\sin\theta)$. Find the coordinates of the point P in each diagram in exact form.

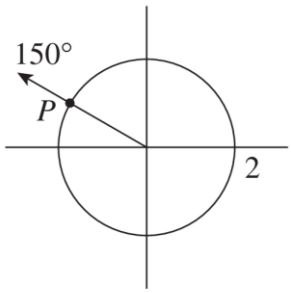
a



$(4\cos 60°, 4\sin 60°)$

$(2, 2\sqrt{3})$

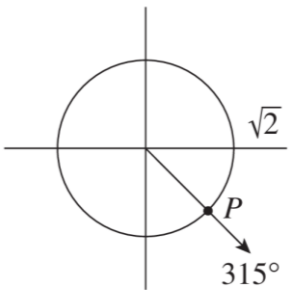
b



$(2\cos 150°, 2\sin 150°)$

$(-\sqrt{3}, 1)$

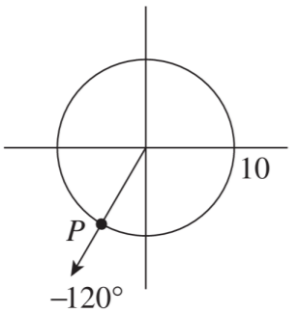
c



$(\sqrt{2}\cos 315°, \sqrt{2}\sin 315°)$

$(1, -1)$

d



$(10\cos -120°, 10\sin -120°)$

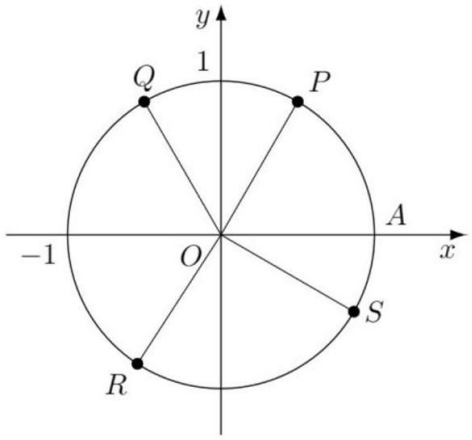
$(-5, -5\sqrt{3})$

9 Find the exact value of:

a	$\tan 225°$	**b**	$\tan 300°$	**c**	$\cos 120°$	**d**	$\cos 330°$
	1		$-\sqrt{3}$		$-\frac{1}{2}$		$\frac{\sqrt{3}}{2}$
e	$\cos(-300°)$	**f**	$\cos(-45°)$	**g**	$\cos 570°$	**h**	$\sin 480°$
	$\frac{1}{2}$		$\frac{1}{\sqrt{2}}$		$-\frac{\sqrt{3}}{2}$		$\frac{\sqrt{3}}{2}$
i	$\sin 300°$	**j**	$\cos(-225°)$	**k**	$\tan(-300°)$	**l**	$\sin(-60°)$
	$-\frac{\sqrt{3}}{2}$		$-\frac{1}{\sqrt{2}}$		$\sqrt{3}$		$-\frac{\sqrt{3}}{2}$
m	$\sin 690°$	**n**	$\sin 495°$	**o**	$\tan 675°$	**p**	$\cos 135°$
	$-\frac{1}{2}$		$\frac{1}{\sqrt{2}}$		-1		$-\frac{1}{\sqrt{2}}$

10 The points P, Q, R, S are on the unit circle. $A(1,0)$ and O is the origin.
Find the exact coordinates of P, Q, R, S if:

$\angle AOP = 60°$ $\angle AOQ = 120°$ $\angle AOR = 240°$ $\angle AOS = 30°$



$P = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), Q = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), R = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right), S = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

11 Prove, using exact values, that:

a $\sin 330° \cos 150° - \cos 390° \sin 390° = 0$

$LHS = -\sin 30°(-\cos 30°) - \cos 30° \sin 30$

$= \sin 30° \cos 30° - \cos 30° \sin 30°$

$= 0 = \text{RHS}$

b $\sin 420° \cos 405° + \cos 420° \sin 405° = \frac{\sqrt{3}+1}{2\sqrt{2}}$

$LHS = \sin 60° \cos 45° + \cos 60° \sin 45°$

$= \frac{\sqrt{3}}{2}\left(\frac{1}{2}\right) + \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)$

$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$

$= \frac{\sqrt{3}+1}{2\sqrt{2}} = RHS$

c $\frac{\sin 135° - \cos 120°}{\sin 135° + \cos 120°} = 3 + 2\sqrt{2}$

$LHS = \frac{\sin 45° - (-\cos 60°)}{\sin 45° + (-\cos 60°)}$

$= \frac{\left(\frac{1}{\sqrt{2}} + \frac{1}{2}\right)}{\left(\frac{1}{\sqrt{2}} - \frac{1}{2}\right)}$

$= \frac{\left(\frac{2+\sqrt{2}}{2\sqrt{2}}\right)}{\left(\frac{2-\sqrt{2}}{2\sqrt{2}}\right)}$

$= \frac{2+\sqrt{2}}{2-\sqrt{2}} \frac{\times 2+\sqrt{2}}{\times 2+\sqrt{2}}$

$= \frac{4+4\sqrt{2}+2}{4-2}$

$= \frac{6+4\sqrt{2}}{2}$

$= 3 + 2\sqrt{2} = RHS$

d $(\sin 150° + \cos 270° + \tan 315°)^2 = \sin^2 135° \cos^2 225°$

$LHS = (\sin 30° + 0 + (-\tan 45°))^2$

$= \left(\frac{1}{2} - 1\right)^2$

$= \frac{1}{4}$

$RHS = \sin^2 45° (- - \cos^2 45°)$

$= \left(\frac{1}{\sqrt{2}}\right)^2 \left(-\frac{1}{\sqrt{2}}\right)^2$

$= \frac{1}{4}$

$\therefore LHS = RHS$

e $\dfrac{\sin 120°}{\tan 300°} - \dfrac{\cos 240°}{\left(\frac{1}{\tan 315°}\right)} = \tan^2 240° - \dfrac{1}{\sin^2 330°}$

$LHS = \dfrac{\sin 60°}{-\tan 60°} - \dfrac{-\cos 60°}{\left(\frac{1}{-\tan 45°}\right)}$

$= \dfrac{\left(\frac{\sqrt{3}}{2}\right)}{-\sqrt{3}} - \dfrac{\left(-\frac{1}{2}\right)}{-1}$

$= -\frac{1}{2} - \frac{1}{2}$

$= -1$

$RHS = \tan^2 60° - \dfrac{1}{\sin^2 30°}$

$= (\sqrt{3})^2 - \dfrac{1}{\left(\frac{1}{2}\right)^2}$

$= 3 - \dfrac{1}{\left(\frac{1}{4}\right)}$

$= 3 - 4$

$= -1$

$\therefore LHS = RHS$

12 Prove that

$$\frac{\cos(90° - \theta)\ \sin(180° - \theta)}{\cos(180° + \theta)\ \cos(-\theta)} = -\tan^2 \theta$$

$LHS = \dfrac{\sin\theta \sin\theta}{-\cos\theta \cos\theta}$

$= \dfrac{\sin^2\theta}{-\cos^2\theta}$

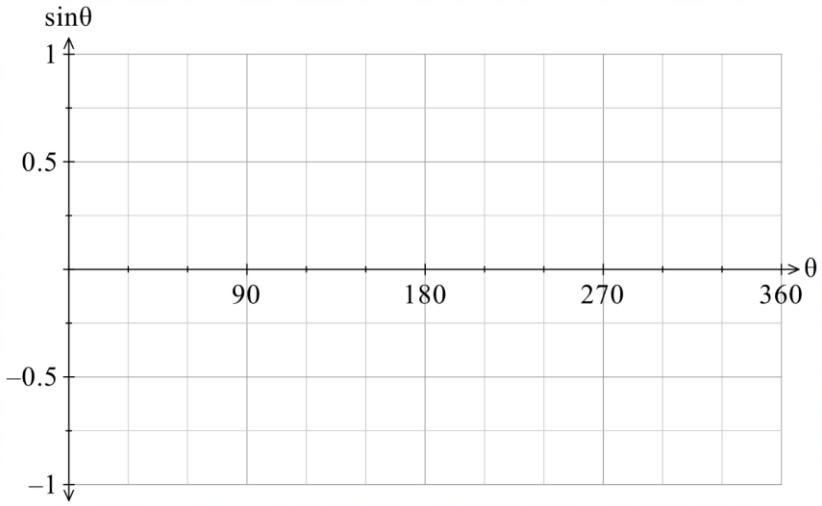
$= -\tan^2\theta = RHS$

Graphs of Trigonometric Functions

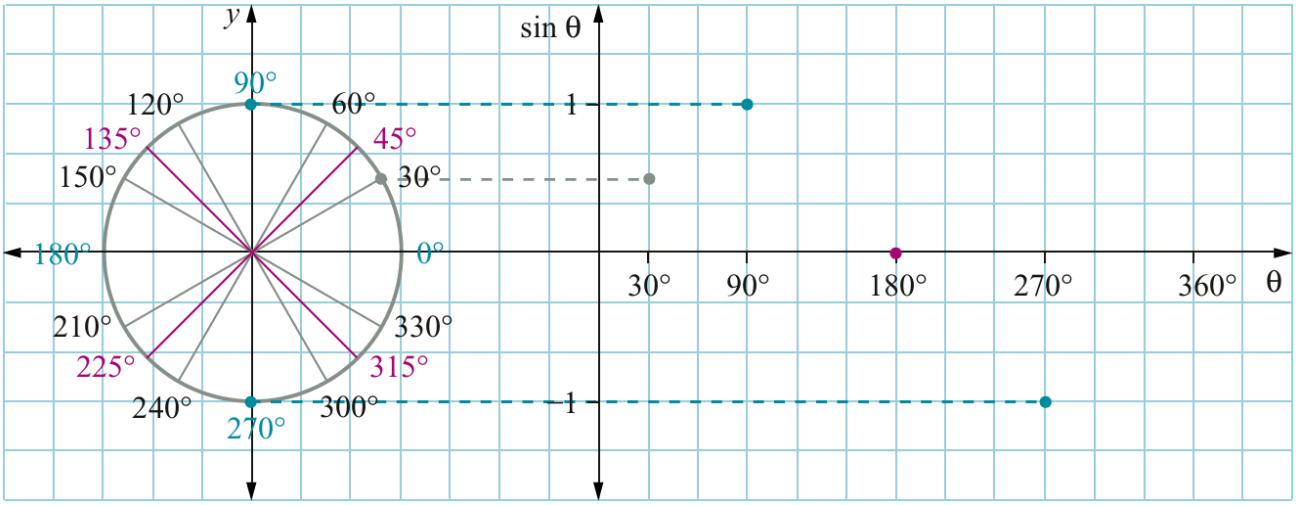
Investigation The graph of $\sin\theta$

The table has been completed for $\sin\theta$ (you can use a calculator to find these values)
Sketch a graph of this table.

$\boldsymbol{\theta}$	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\boldsymbol{\sin\theta}$	0	0.5	0.87	1	0.87	0.5	0	−0.5	−0.87	−1	−0.87	−0.5	0



We can also graph the sine function by considering the y coordinates on a unit circle.



Properties of the sine curve $\sin\theta$:

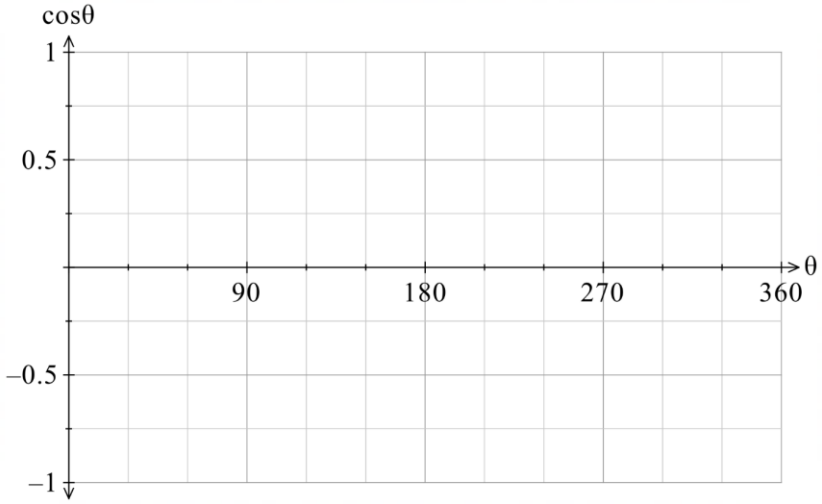
- The $\sin\theta$ value **oscillates** (swings repeatedly) between …… and ……
- The **period** (time to complete one cycle) is ……°, so the graph repeats every ……°
- In $0° < \theta < 180°$, the value of $\sin\theta$ is *positive* | *negative*, confirming the ASTC rule that sine is positive in quadrants …… and ……
- In $180° < \theta < 360°$, the value of $\sin\theta$ is *positive* | *negative*, confirming the ASTC rule that sine is negative in quadrants …… and ……
- In $0° \leq \theta \leq 360°$, when $\theta =$ ……, ……, and ……, the value of $\sin\theta$ is ……

Investigation The graph of $\cos\theta$

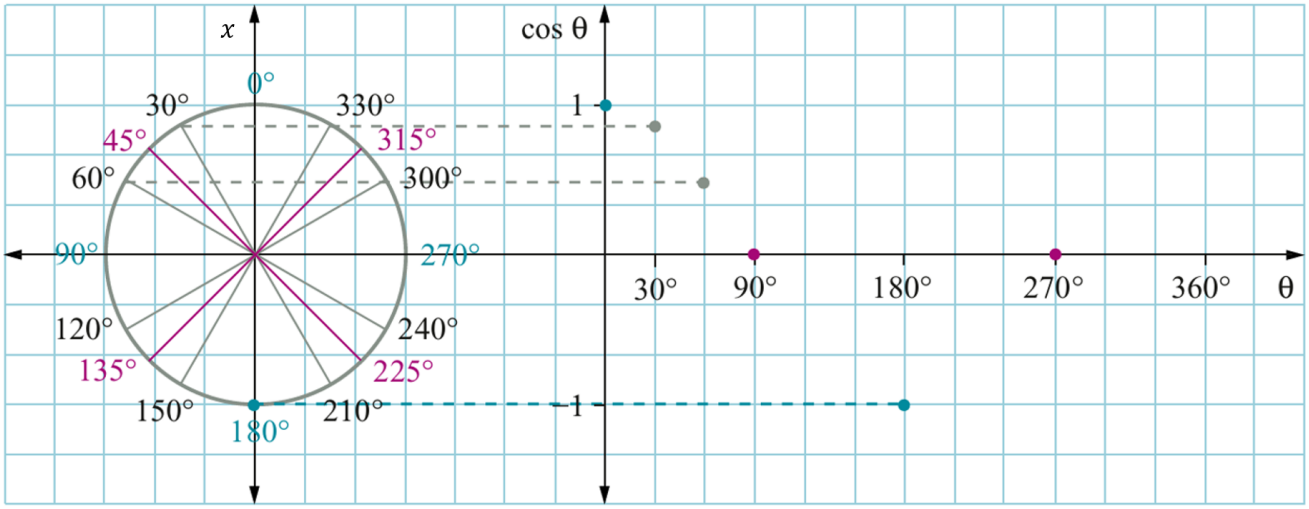
The table has been completed for $\cos\theta$

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\cos\theta$	1	0.87	0.5	0	-0.5	-0.87	-1	-0.87	-0.5	0	0.5	0.87	1

Sketch a graph of this table.



Graph $\cos\theta$ by considering the x **coordinates** on a unit circle.



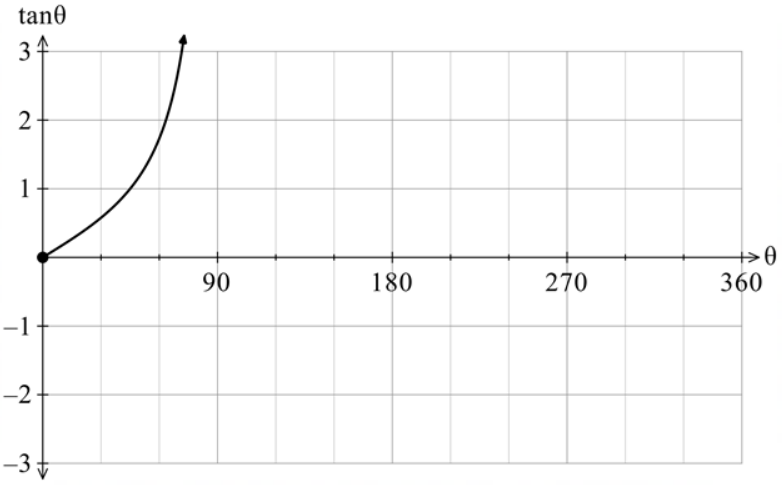
Properties of the cosine curve $\cos\theta$:

- The $\cos\theta$ value **oscillates** (swings repeatedly) between …… and ……
- The **period** (time to complete one cycle) is ……°, so the graph repeats every ……°
- In $0° < \theta < 90°$, the value of $\cos\theta$ is *positive* | *negative*, confirming the ASTC rule that cos is positive in quadrant ……
- In $90° < \theta < 270°$, the value of $\cos\theta$ is *positive* | *negative*, confirming the ASTC rule that cos is negative in quadrants …… and ……
- In $270° < \theta < 360°$, the value of $\cos\theta$ is *positive* | *negative*, confirming the ASTC rule that cos is positive in quadrant ……
- In $0° \leq \theta \leq 360°$, when $\theta =$ …… and …… the value of $\cos\theta$ is ……
- Comparing the circle diagrams for $\sin\theta$ and $\cos\theta$, the circle diagram for $\cos\theta$ has been rotated ……-wise by ……°, as a result, the graph of $\cos\theta$ is the graph of $\sin\theta$ shifted by ……°. This agrees the with the identity $\cos\theta = \sin(……° - \theta)$

Investigation The graph of $\tan\theta$

We will do the same thing for $\tan\theta$:

$\boldsymbol{\theta}$	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\boldsymbol{\tan\theta}$	0	0.58	1.73	und	−1.73	−0.58	0	0.58	1.7	und	−1.7	−0.58	0



Properties of the tan curve $\tan\theta$:

- The $\tan\theta$ value has …… maximum nor minimum.
- The **period** (time to complete one cycle) is ……°, so the graph repeats every ……°
- $\tan\theta$ is positive when …… $< \theta <$ …… and …… $< \theta <$ ……, confirming the ASTC rule that tan is positive in quadrants …… and ……
- $\tan\theta$ is negative when …… $< \theta <$ …… and …… $< \theta <$ ……, confirming the ASTC rule that tan is negative in quadrants …… and ……
- There are vertical …… at $\theta = 90°$ and $270°$. These are values where $\cos\theta = 0$.
- These asymptotes occur because $\tan\theta = \frac{\sin\theta}{\cos\theta}$; when $\cos\theta = 0$, dividing by 0 is ……

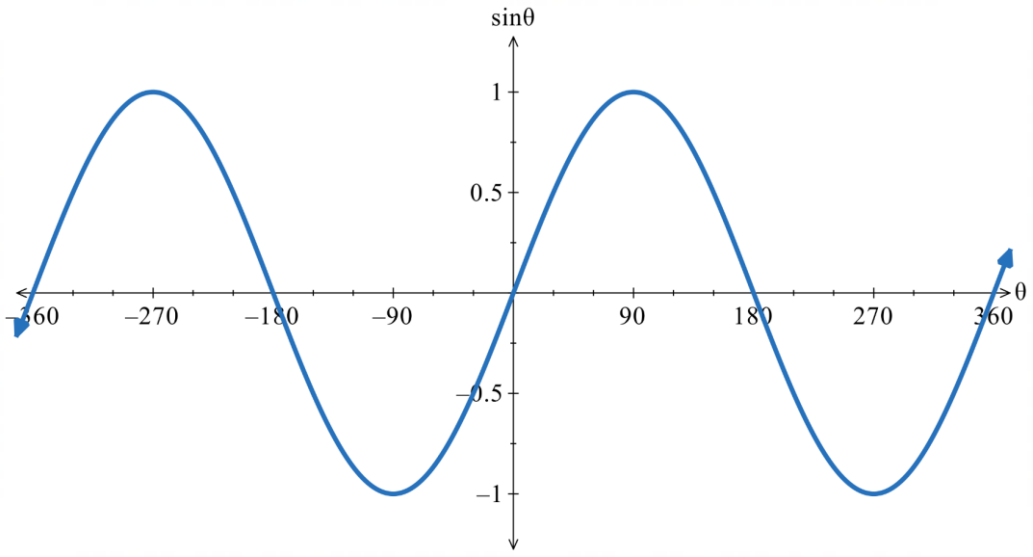
ⓘ Graphs of the Trigonometric Functions

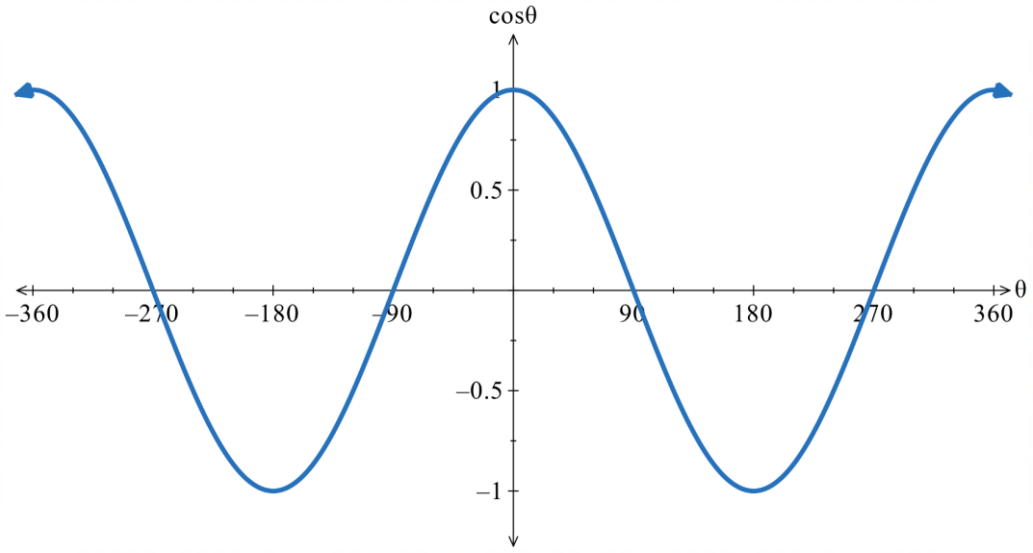
The graphs of the trigonometric functions are **periodic**, they keep repeating.

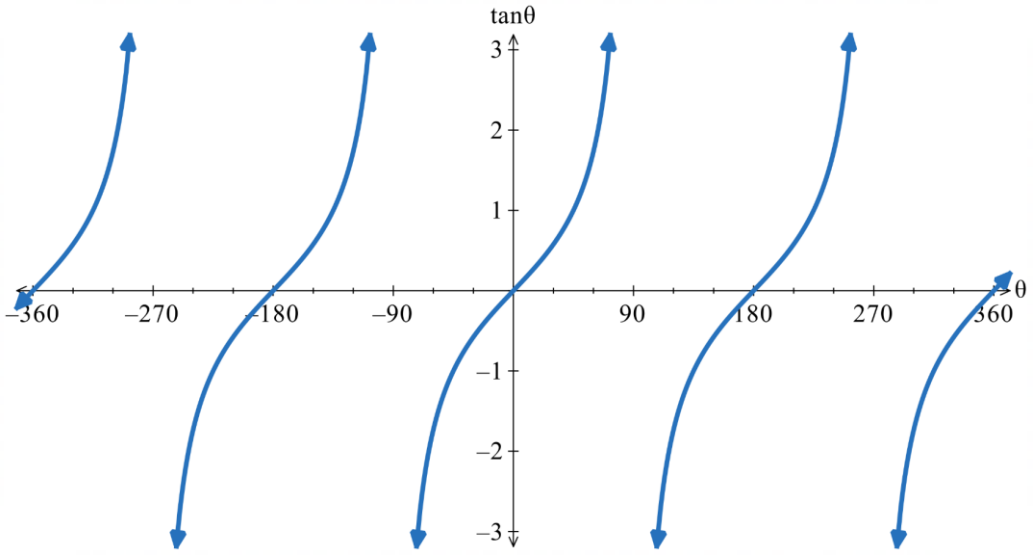
$\cos x$ is $\sin x$ shifted by $90°$: $\cos x = \sin(90 - x)$.

$\sin x$ and $\cos x$ are called **sinusoidal**, they occur when you analyse waves.

https://www.geogebra.org/m/mynd2psx







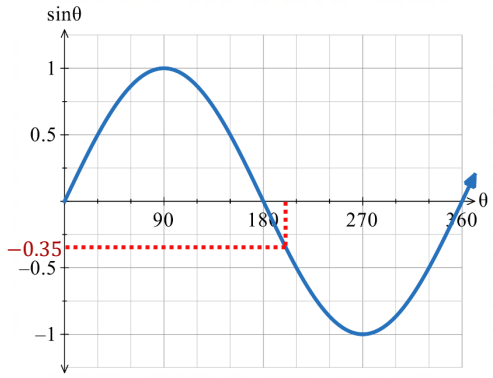
Using the Trigonometric Graphs to Evaluate and Solve Equations

We can use the trigonometric graphs to estimate values of $\sin\theta$ and solve equations.

They are also useful because we can quickly determine how many solutions there are in a given interval of θ.

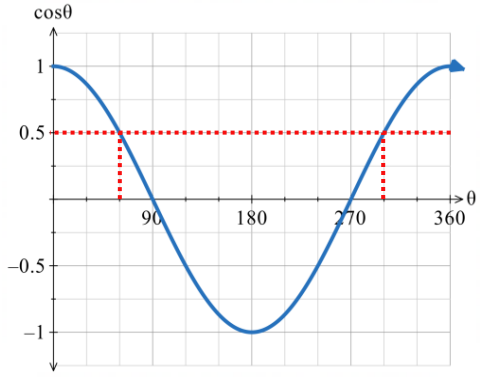
Example Interpret graphs of trigonometric functions

Estimate $\sin 200°$ using the graph



$\sin 200° \approx -0.35$

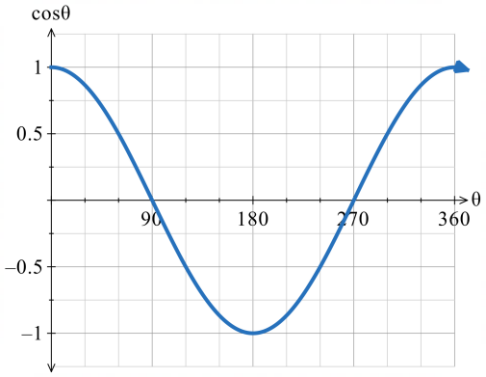
Find values of θ in $0 \leq \theta < 360°$ where $\cos\theta = 0.5$ using the graph.



$\theta = 60°$ or $300°$

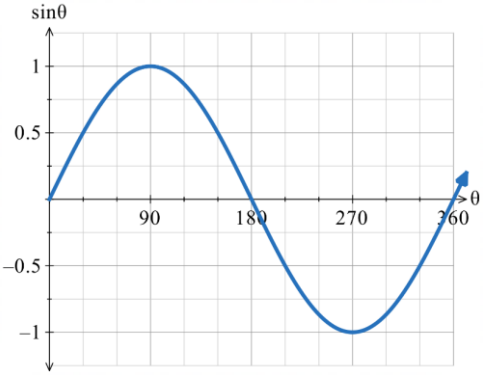
Guided Practice

a Use the graph to estimate $\cos 100°$



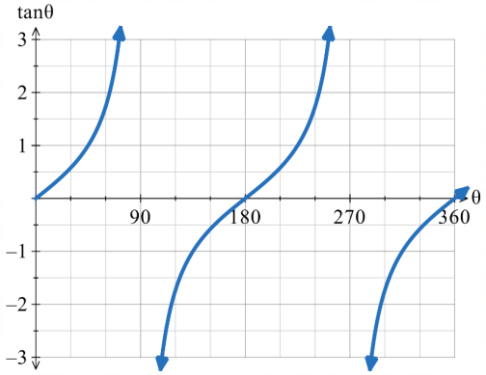
≈ -0.2

b Find values of θ in $0 \leq \theta < 360°$ where $\sin\theta = -0.5$ using the graph.



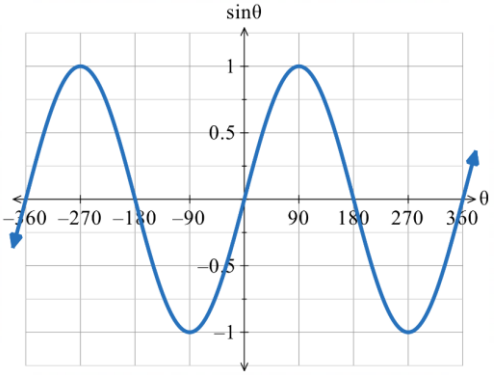
$210°, 330°$

c Find values of θ in $0 \leq \theta < 360°$ where $\tan\theta = 2$ using the graph.



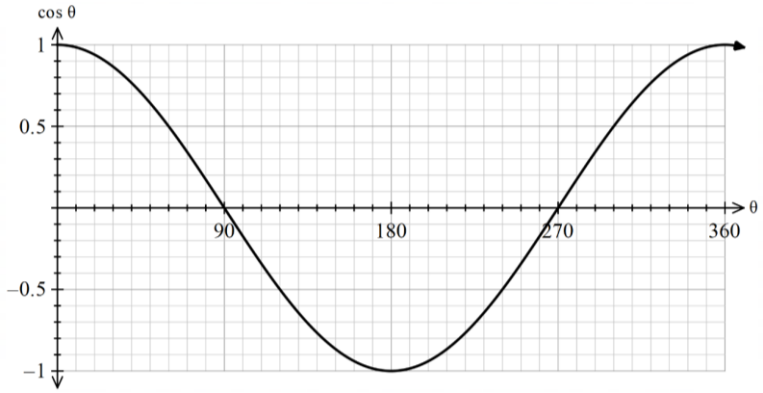
$\approx 60°, 240°$

d How many values of θ in $-360° < \theta < 360°$ satisfy $\sin\theta = 0.75$?



4

1 This graph shows $\cos\theta$. Use it to estimate the following to 2 d.p.

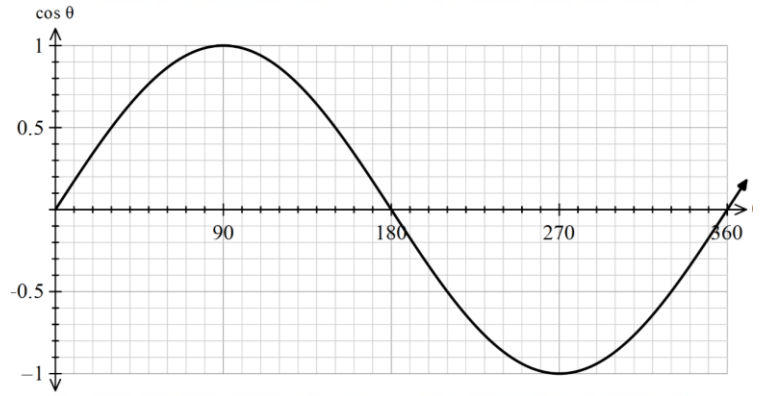


a $\cos 35°$	**b** $\cos 190°$	**c** $\cos 330°$	**d** $\cos 140°$
0.82	− 0.98	0.87	− 0.77

2 Use the same graph as in Q1 to estimate the **values** of θ between 0° and 360° for:

a $\cos\theta = 0.8$	**b** $\cos\theta = 0.6$	**c** $\cos\theta = 0$	**d** $\cos\theta = -0.4$
37°, 323°	53°, 307°	90°, 270°	114°, 246°

3 Use the graph of $\sin\theta$ to evaluate/solve:

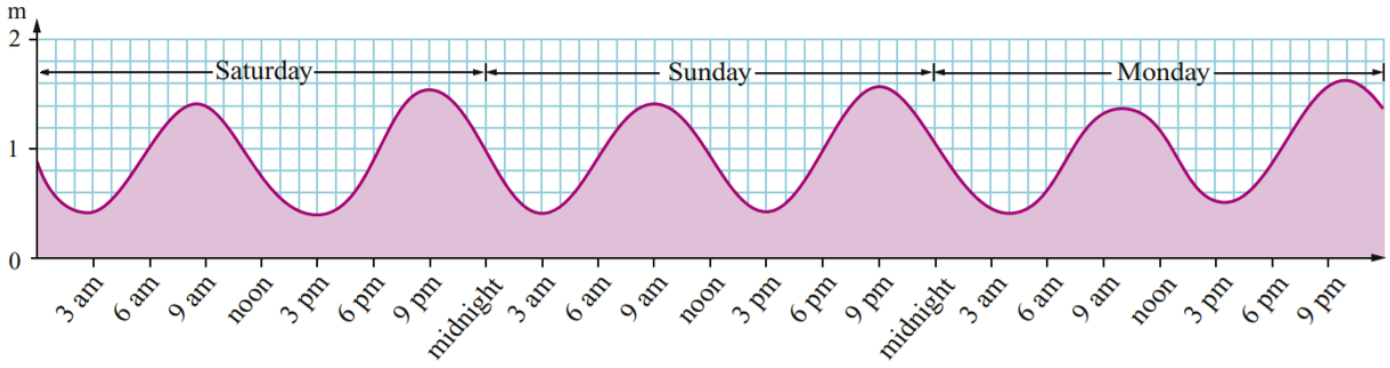


a $\sin 25°$	**b** $\sin 115°$	**c** $\sin 220°$	**d** $\sin 310°$
0.42	0.91	− 0.64	− 0.77
e $\sin\theta = 0.6$	**f** $\sin\theta = 0.2$	**g** $\sin\theta = -0.4$	**h** $\sin\theta = -0.7$
37°, 143°	12°, 168°	204°, 336°	233°, 307°

4 Use the trigonometric graphs to find the values (if they exist) of these trigonometric ratios of boundary angles.

a $\sin 0°$	**b** $\cos 180°$	**c** $\cos 90°$	**d** $\tan 0°$
0	−1	0	0
e $\sin 90°$	**f** $\cos 0°$	**g** $\sin 270°$	**h** $\cos 270°$
1	1	−1	0
i $\sin 180°$	**j** $\tan 270°$	**k** $\tan 90°$	**l** $\tan 180°$
0	undefined	undefined	0

5 A tidal chart is shown below.



a How many low tides occur in a day?

2

b How many high tides occur in a day?

2

c What word is used to describe graphs of this shape?

sinusoidal

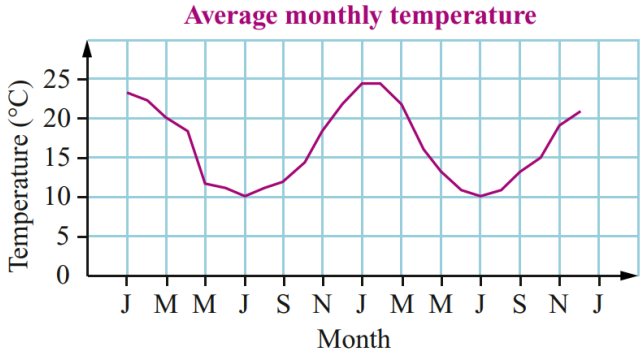
d Estimate the low tide height and high tide height for Sunday between noon and midnight.

low tide: 0.4 m, high tide: 1.6 m

e Estimate the period for the tide graph.

Around 12 hours

6 The graph shows the average monthly temperature of a city over 2 years, starting in Jan.



a Would you expect this city to be in the Northern or Southern Hemisphere?

Southern, temperatures are high in January.

b Which trigonometric graph does this resemble? A cosine or sine graph?

Cosine

c If temperatures of a city in the Northern hemisphere were graphed, what graph would you expect, and why?

Sine; temperatures would be cold in January.

d What is the difference between the highest and lowest values?

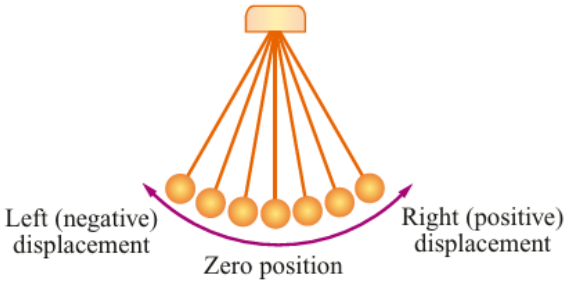
Around 14°C

e What is the period of this graph?

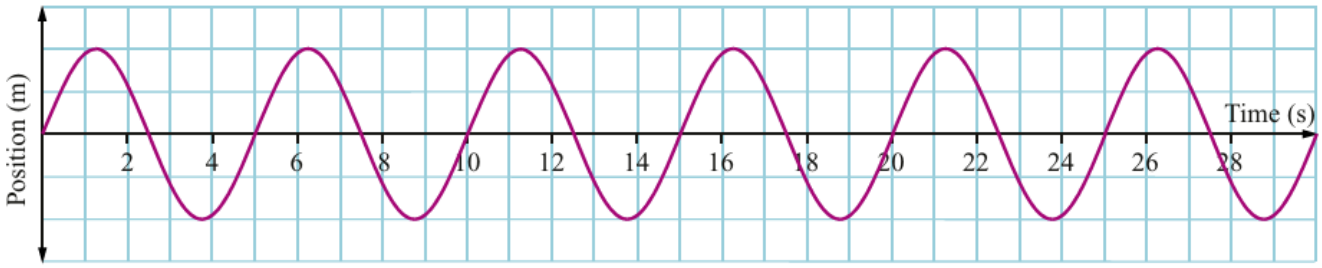
1 year

7 The diagram shows the motion of a swinging pendulum and its graph.

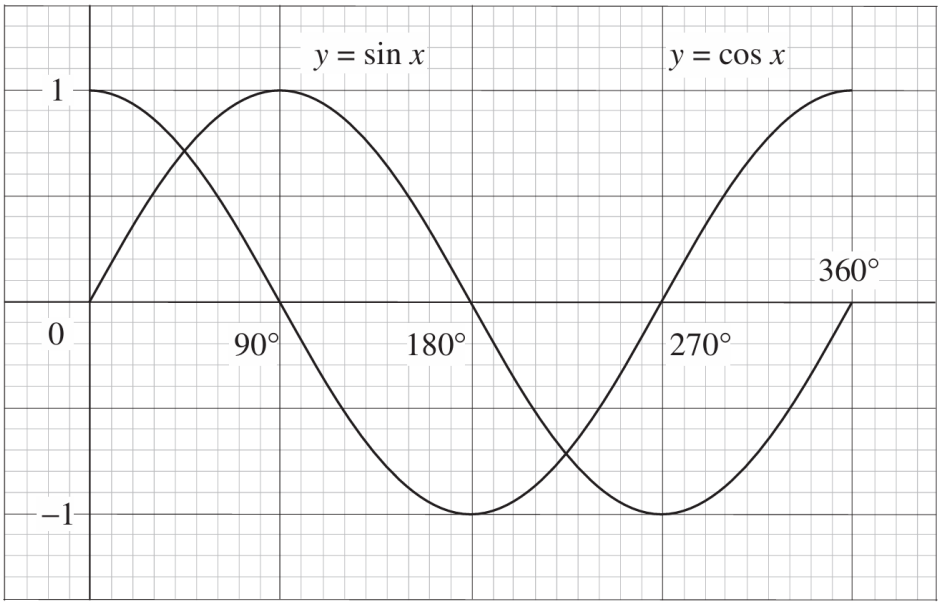
What is the period if this graph?



5 seconds



8 Consider the graphs of $y = \cos x$ and $y = \sin x$ on the same axis:



Decide whether the following statements are true or false.

a	sin 60° > sin 200° T	**b**	sin 100° < sin 300° F	**c**	sin 135° < sin 10° F
d	sin 200° = sin 340° T	**e**	cos 70° < cos 125° F	**f**	cos 315° > cos 135° T
g	cos 310° = cos 50° T	**h**	cos 95° > cos 260° T	**i**	sin 90° = cos 360° T
j	cos 180° = sin 180° F	**k**	sin 210° > sin 285° T	**l**	cos 15° >cos 115° T

9 Using the graph in the previous question, for what angles are the values of $\sin x$ and $\cos x$ equal in the domain $0° \leq x < 360°$?

45° and 135°

Odd and Evenness of Trigonometric Functions

ⓘ Symmetry of Trigonometric Functions

sin x and **tan x** are **odd**		cos x is **even**
$\sin(-x)$	$\tan(-x)$	$\cos(-x)$
$= -\sin x$	$= -\tan x$	$= \cos x$

FOUNDATION

1 What is the domain and range of $y = \sin\theta$?

D: all real θ

R: $[-1,1]$

2 What is the domain and range of $y = \cos\theta$?

D: all real θ

R: $[-1,1]$

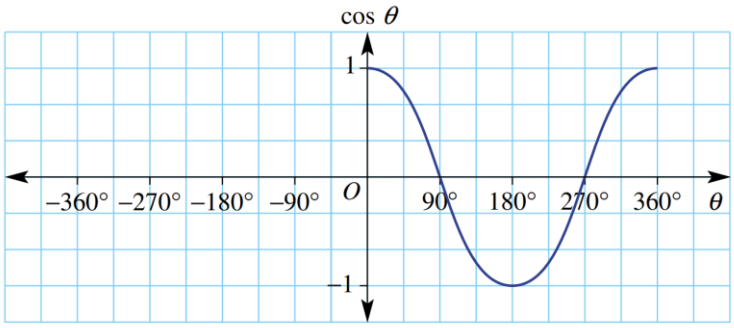
3 Explain why no values of θ satisfy sin θ = 2.

No solution as 2 is not in the range of $y = \sin\theta$.

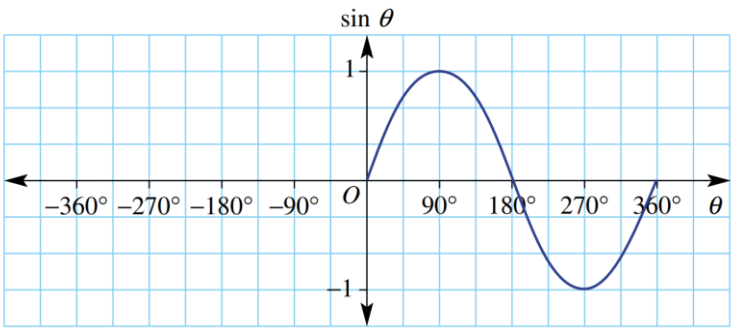
4 How many values of θ satisfy cos θ = −4? Give a reason.

None. -4 is not in the range of $y = \cos\theta$.

5 $y = \cos\theta$ is an even function as $\cos(-\theta) = \cos\theta$.
Complete the graph for the domain $-360° \leq \theta \leq 360°$



6 $y = \sin\theta$ is an odd function as $\sin(-\theta) = -\sin\theta$.
Complete the graph for the domain $-360° \leq \theta \leq 360°$



7 If $\sin x = 0.8$, find $\sin(-x)$.

-0.8

8 If $\tan(-x) = \sqrt{7}$, find the value of $-\tan x$.

$\sqrt{7}$

9 If $\cos(-x) = 0.3$, find the value of $\cos x$.

0.3

10 Simplify $\tan(-\alpha) - \tan(\alpha)$.

$-2\tan\alpha$

11 Simplify $\cos(-\theta) - \cos\theta$.

0

12 $f(x) = \sin x + \cos(90° - x)$. Decide whether $f(x)$ is odd, even, or neither.

$$f(x) = \sin x + \sin x = 2\sin x$$
$$f(-x) = 2\sin(-x)$$
$$f(-x) = -2\sin x$$
$$= -f(x)$$
$$\therefore \text{odd}$$

13 $g(x) = x\sin(-x) + \cos(-x)$. Decide whether $g(x)$ is odd, even, or neither.

$$g(x) = -x\sin x + \cos x$$
$$g(-x) = -(-x)\sin(-x) + \cos(-x)$$
$$g(-x) = -x\sin x + \cos x$$
$$= g(x)$$
$$\therefore \text{even}$$

Sketch this on Desmos, it looks really cool!

14 $f(x) = x^2\sin(-x) + x\cos(-x)$. Decide whether $f(x)$ is odd, even, or neither.

$$f(x) = -x^2\sin x + x\cos x$$
$$f(-x) = -(-x)^2\sin(-x) + (-x)\cos(-x)$$
$$= x^2\sin x - x\cos x$$
$$= -f(x)$$
$$\therefore \text{odd}$$

Sketch this on Desmos, it looks really cool!

Solving Trigonometric Equations

Investigation Solving trigonometric equations

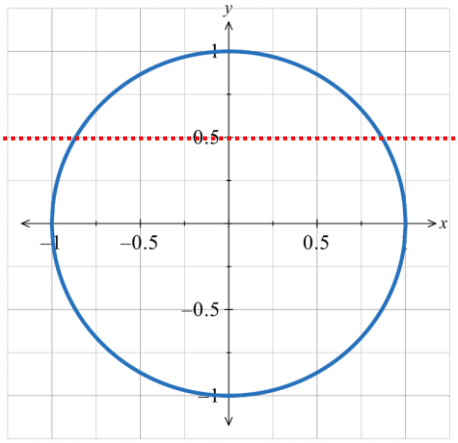
Consider $\sin\theta = 0.5$. To solve this equation, we can use the inverse trigonometric function:

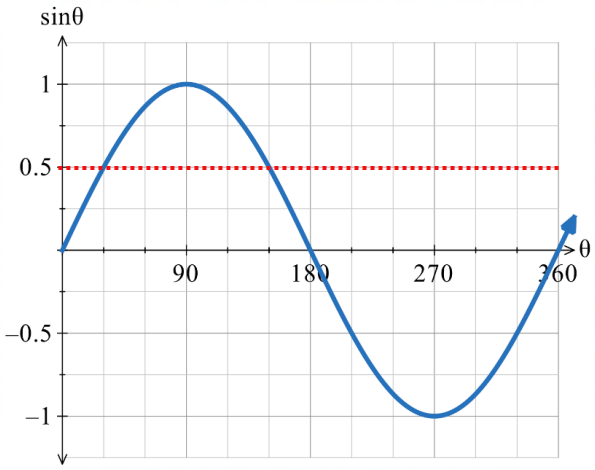
$$\sin\theta = 0.5$$

$$\theta = \ldots\ldots\ldots\ldots$$

The calculator will return only one value: $\sin^{-1}(0.5) = 30°$.

However, using the unit circle, or the graph, there are **two solutions** for $0 \leq \theta < 360°$.





To solve trigonometric equations without sketching an accurate unit circle and graph, we need another method.

It involves finding the **acute reference angle** and **identifying the relevant quadrants**.

ⓘ Finding the Reference Angle from a Calculator

To find the acute reference angle A:

- Use the relevant inverse trigonometric function: $\sin^{-1}(\ ), \cos^{-1}(\ ), \tan^{-1}(\ )$
- Always enter the **positive** value of the ratio.
- The calculator returns the acute reference angle A.

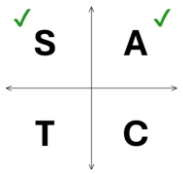
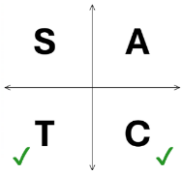
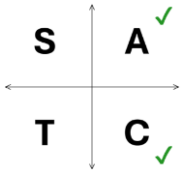
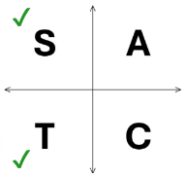
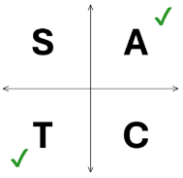
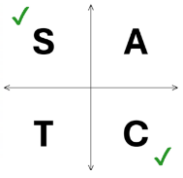
The calculator's inverse functions give results in restricted ranges that may include negative angles.
By entering the positive value, we always get a positive acute angle and this leads to fewer mistakes.
You will learn more about the inverse trigonometric functions in the Year 12 Advanced course.

🔍 **Determine** reference angle from a calculator

$\sin\theta = \frac{1}{\sqrt{2}}$ $A = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ $= 45°$	$\cos\theta = -\frac{\sqrt{3}}{2}$ $A = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ $= 30°$	$\tan\theta = -0.5$ $A = \tan^{-1}(0.5)$ $\approx 26°34'$
Find the acute reference angle for these trigonometric ratios.		
a $\sin\theta = -\frac{1}{2}$	**b** $\tan\theta = -1$	**c** $\cos\theta = \frac{1}{\sqrt{2}}$
d $\tan\theta = \sqrt{3}$	**e** $\tan\theta = -\frac{1}{\sqrt{3}}$	**f** $\sin\theta = -0.6$

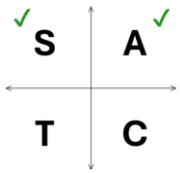
ⓘ Identifying the Quadrant from a Trigonometric Ratio

We identify the quadrants from the sign of the trigonometric ratio:

Condition	$\sin\theta > 0$	$\sin\theta < 0$	$\cos\theta > 0$	$\cos\theta < 0$	$\tan\theta > 0$	$\tan\theta < 0$
Quadrants	Q1, Q2	Q3, Q4	Q1, Q4	Q2, Q3	Q1, Q3	Q2, Q4
Diagram						

🔍 **Determine** quadrants from a ratio

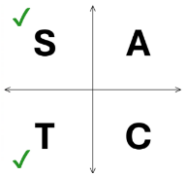
$\sin\theta = \frac{1}{\sqrt{2}}$



$\sin\theta > 0$

$\therefore Q1, Q2$

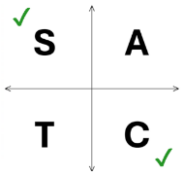
$\cos\theta = -\frac{\sqrt{3}}{2}$



$\cos\theta < 0;$

$\therefore Q2, Q3$

$\tan\theta = -0.5$

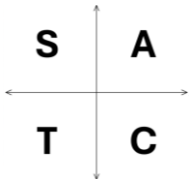


$\tan\theta < 0;$

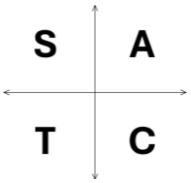
$\therefore Q2, Q4$

Find the acute reference angle for these trigonometric ratios.

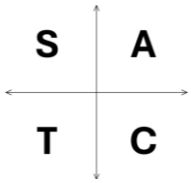
a $\sin\theta = -\frac{1}{2}$



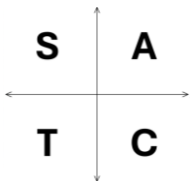
b $\tan\theta = -1$



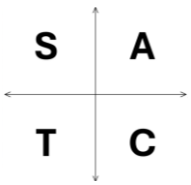
c $\cos\theta = \frac{1}{\sqrt{2}}$



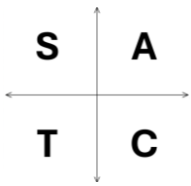
d $\tan\theta = \sqrt{3}$



e $\tan\theta = -\frac{1}{\sqrt{3}}$



f $\sin\theta = -0.6$



Solving Trigonometric Equations

To solve equations like $\text{trig}\,\theta = \text{value}$:

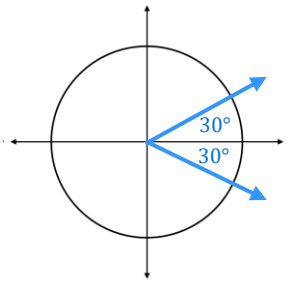
1. Find the acute reference angle using the inverse function on the calculator.
   Remember to always use the positive ratio. $A = \text{trig}^{-1}(|\text{value}|)$
2. Determine the quadrants using ASTC based on the sign of the value.
3. Calculate θ for each quadrant.

⚠ For now, you will only be asked to solve equations where $0 \leq \theta < 360°$

Example Solve simple trigonometric equations where $0 \leq \theta < 360°$

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 360°$

1. $A = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 30°$
2. $\cos x > 0$,
   θ is in Q1 or Q4.
3. $\theta = 30°$ or $330°$



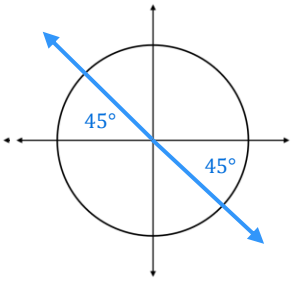
How did we know the second solution is $330°$?

Since cos is also positive in the …th quadrant

The reference angle of $30°$ in the …th quadrant represents an angle of inclination of …… $-30° = 330°$

Solve $\tan\theta = -1$ for $0 \leq \theta < 360°$

1. $A = \tan^{-1}(1) = 45°$
2. $\tan\theta < 0$,
   θ is in Q2 or Q4.
3. $\theta = 135°$ or $315°$



Why is $45°$ not a solution?

The question says $\tan\theta = -1$, and *tan is …… in the first quadrant.*

Type in $\tan^{-1}(-1)$ into the calculator.
Why do we always use the positive ratio?

By using the positive ratio, the calculator always gives us the …… angle.

Guided Practice

Solve these trigonometric equations for $0 \leq \theta < 360°$

a $\cos\theta = \frac{1}{\sqrt{2}}$

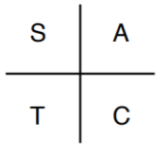
$45°, 315°$

b $\sin\theta = -\frac{\sqrt{3}}{2}$

$240°, 300°$

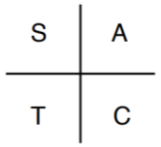
1 For each equation, circle the quadrants that x would be in.

a $\sin x = \frac{1}{2}$



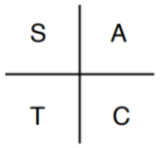
A S

b $\sin x = -\frac{1}{2}$



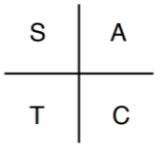
T C

c $\tan x = 3$



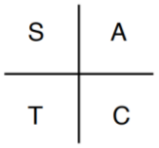
A T

d $\tan x = -3$



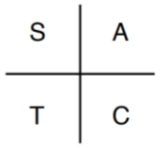
S C

e $\cos x = \frac{2}{5}$



A C

f $\cos x = -\frac{2}{5}$



S T

2 Solve each of these equations for $0 \leq \theta < 360°$

a $\sin\theta = \frac{\sqrt{3}}{2}$

1. Reference angle

2. Quadrants

3. Solutions

θ =60° or 120°

b $\tan\theta = 1$

θ =45° or 225°

c $\tan\theta = \sqrt{3}$

θ =60° or 240°

d $\cos\theta = -\frac{1}{\sqrt{2}}$

θ =135° or 225°

e $\tan\theta = -\sqrt{3}$

θ =120° or 300°

f $\cos\theta = -\frac{\sqrt{3}}{2}$

θ =150° or 210°

3 These equations lie on boundary angles, i.e. they are on the edges of the quadrants.
We solve these using the graphs, a diagram of the unit circle, or knowledge of exact values.
Solve for $0 \leq \theta < 360°$

a $\sin\theta = 1$

θ =90°

b $\cos\theta = 1$

θ = 0°

c $\tan\theta = 0$

θ = 0°, 180°

d $\sin\theta = -1$

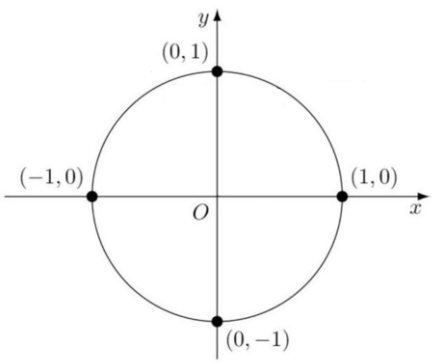
θ = 270°

e $\cos\theta = 0$

θ = 90° or 270°

f $\sin\theta = 0$

θ = 0°, 180°



4 Solve each of these equations for $0 \leq x < 360°$ to the nearest degree.
You will need to enter the inverse trigonometric function into a calculator.

a $\cos x = \frac{3}{7}$

65° or 295°

b $\sin x = 0.1234$

7° or 173°

c $\tan x = 7$

82° or 262°

d $\sin x = -\frac{2}{3}$

222° or 318°

e $\tan x = -\frac{20}{9}$

114° or 294°

f $\cos x = -0.77$

140° or 220°

g $\tan x = -0.3$

163° or 343°

h $\cos x = 0$

90° or 270°

i $\sin x = -\frac{\sqrt{3}}{2}$

240° or 300°

j $\cos x = 0.191$

79° or 281°

k $\tan x = 1$

45° or 225°

l $\frac{1}{\cos x} = \sqrt{2}$

45° or 315°

5 Solve for $\theta \in [0°, 360°)$.

a $2\sin\theta - 1 = 0$

30°, 150°

b $\sqrt{2}\cos\theta - 1 = 0$

45°, 315°

c $\sqrt{3}\tan\theta = -2 = -1$

30°, 210°

d $2\cos\theta + \sqrt{3} = 0$

150°, 210°

e $6\sin\theta + 3 = 0$

210°, 330°

f $1 + \tan\theta = 0$

135°, 315°

6 Solve for $\theta \in [0°, 360°)$. Remember to take positive and negative values when square rooting.

a $\sin^2\theta = 1$

90°, 270°

b $\cos^2\theta = \frac{3}{4}$

30°, 150°, 210°, 330°

c $\tan^2\theta = 3$

60°, 120°, 240°, 300°

7 Solve for $\theta \in [0°, 360°)$. Remember, to solve quadratic equations we need to factorise.

a $2\cos^2\theta - \cos\theta = 0$

60°, 90°, 270°, 300°

b $\sin^2\theta + \sin\theta = 0$

0°, 180°, 270°

c $\tan^2\theta = \tan\theta$

0°, 45°, 180°, 225°

d $2\cos^2\theta - 1 = 0$

45°, 135°, 225°, 315°

e $3\tan^2 - 1 = 0$

30°, 150°, 210°, 330°

f $(2\sin\theta - 1)(\sin\theta + 1) = 0$

30°, 150°, 270°

8 Jimmy solved $\tan^2\theta = \tan\theta$ for $\theta \in [0°, 360°)$ by dividing both sides by $\tan\theta$ first.
He got the solutions $45°$ and $225°$, which are not all the solutions.

The correct method is to move all terms to left side and factorise, then use null factor law.

By dividing both sides by $\tan\theta$, he is assuming that $\tan\theta$ is not 0.

Therefore, he is missing the solutions to $\tan\theta = 0$, which are $\theta = 0°$ and $\theta = 180°$

9 Solve for $\theta \in [0°, 360°)$.

a $2\sin^2\theta - 3\sin\theta + 1 = 0$

$30°, 90°, 150°$

b $2\cos^2\theta - \cos\theta - 3 = 0$

$180°$

c $2\cos^2\theta = \cos\theta$

$60°, 90°, 270°, 300°$

d $\frac{1}{\cos^2\theta} + \frac{1}{\cos\theta} - 2 = 0$

$0°, 120°, 240°, 360°$

10 Solve for $\theta \in [0°, 360°)$. Answer to the nearest minute where necessary.

a $2\sin\theta\cos\theta = \sin\theta$

$0°, 60°, 180°, 300°, 360°$

b $2\sin^2\theta + \sin\theta = 1$

$30°, 150°, 270°$

c $\frac{1}{\cos^2\theta} + \frac{2}{\cos\theta} = 8$

$60°, 300°, 104°29', 255°, 31'$

d $3\cos^2\theta + 5\cos\theta = 2$

$70°32', 289°28'$

e $\frac{4}{\sin^2\theta} - \frac{4}{\sin\theta} - 15 = 0$

$23°35', 156°25', 22°\,49', 31°\,11'$

f $4\sin^3\theta = 3\sin\theta$

$0°, 60°, 120°, 180°, 240°, 300°, 360°$

11 Solve $4\cos^2\theta + 2\sin\theta = 3$ for $0 \leq x < 360°$ to the nearest degree.

$\sin\theta = \frac{1 \pm \sqrt{5}}{4}$

$\theta = 54°, 126°, 198°, 342°$

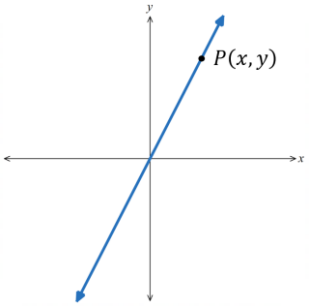
Gradient and Angle of Inclination

Investigation Gradient and Tan

Consider a line through the origin. Choose any point (x, y) on the line.

The gradient is: $m = \frac{\text{rise}}{\text{run}} =$

Construct a right-angled triangle, the ratio $\tan\theta = \frac{\text{opp}}{\text{adj}} =$



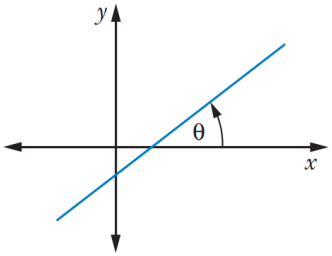
Therefore, $m = \tan\theta$.

We can extend this relationship to all lines, not just ones through the origin, since the gradient and angle of inclination does not change based on the location of the line.

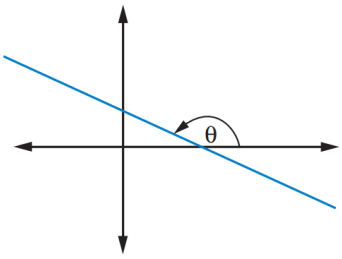
Gradient is Tan of Angle of Inclination

$$m = \tan\theta$$

For acute angles, the gradient is positive.

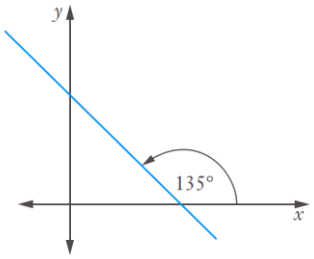


For obtuse angles, the gradient is negative.



Example Determine gradient of a line from the angle of inclination

Find the gradient of the line that makes an angle of inclination of $135°$



$$m = \tan\theta$$
$$= \tan 135°$$
$$= -1$$

Guided Practice

a $\theta = 45°$

1

b $\theta = 80°$

5.67

c $\theta = 155°$

−0.47

d $\theta = -45°$

−1

ⓘ Finding the Angle of Inclination from the Gradient

$$\theta = \tan^{-1}(m)$$

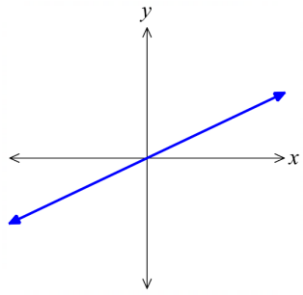
If the calculator returns a negative angle, add 180° to the result.

⚠ $\tan^{-1}(\ \ )$ function always returns angles in the range $-90° < \theta < 90°$.
Angle of inclination of a line is an angle $0 \leq \theta < 180°$.
We add 180° to the result because $\tan\theta = \tan(180° + \theta)$.

Example Determine angle of inclination from the gradient

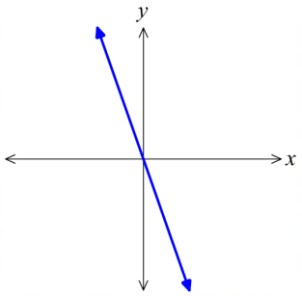
Find, correct to the nearest minute, the angle of inclination of a straight line with gradient:

$m = 0.5$



$$\tan\theta = m$$
$$\tan\theta = 0.5$$
$$\theta = \tan^{-1}(0.5)$$
$$\theta \approx 26°34'$$

$m = -3$

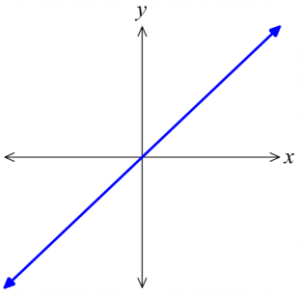


$$\tan\theta = m$$
$$\tan\theta = -3$$
$$\theta = \tan^{-1}(-3)$$
$$\theta \approx -71°35'$$
$$\approx 108°26'$$

Guided Practice

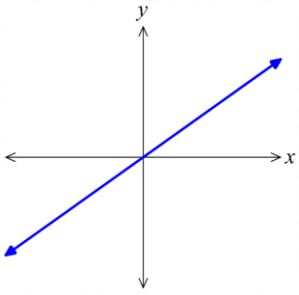
Find, correct to the nearest minute, the angle of inclination of a straight line with gradient:

a $m = 1$



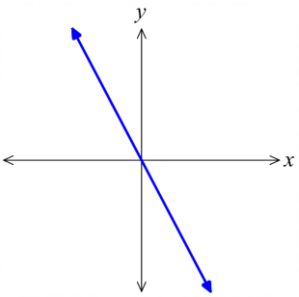
45°

b $m = \frac{3}{4}$



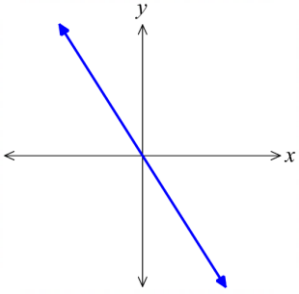
36° 52′

c $m = -2$



116°34′

d $m = -\frac{5}{3}$



300°58′

1 Use the formula $m = \tan\theta$ to find the gradient of a line with these angles of inclination. Answer to 2 decimal places.

a $\theta = 15°$

0.27

b $\theta = 85°$

11.43

c $\theta = 135°$

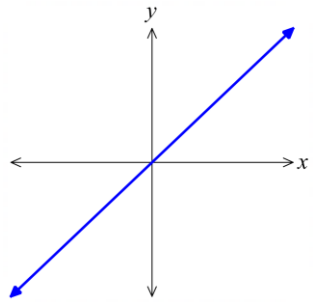
–1

d $\theta = 180°$

0

2 Use the formula $\tan\theta = m$ to find the angle of inclination of a line with these gradients. Answer to the nearest degree. A sketch will help you visualise the problem.

a $m = 1$



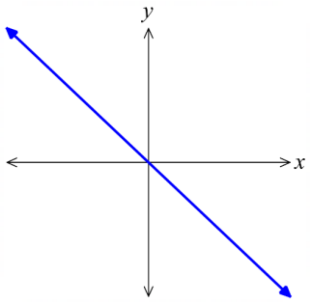
$\tan\theta = 1$

$\theta = \tan^{-1}(\quad)$

$=$

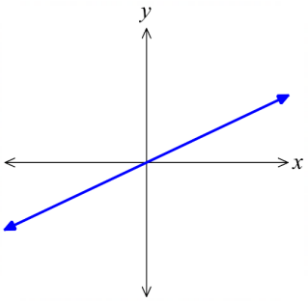
45°

b $m = -1$



135°

c $m = 0.5$



27°

d $m = -0.5$

153°

e $m = 1.73$

60°

f $m = \frac{5}{2}$

68°

g $m = 4$

76°

h $m = -0.58$

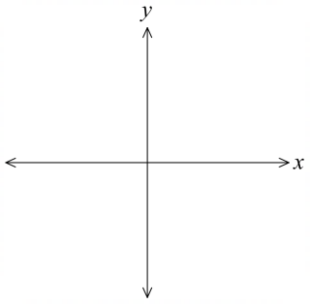
150°

i $m = -12$

95°

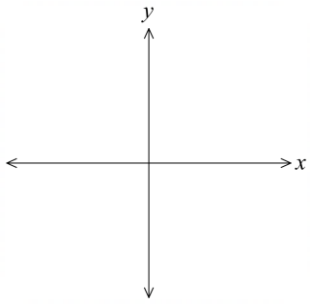
3 Sketch these lines then find the angle of inclination to the nearest degree.

a $y = 3x + 6$



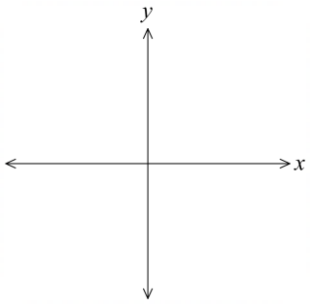
72°

b $y = -\frac{1}{2}x + 1$



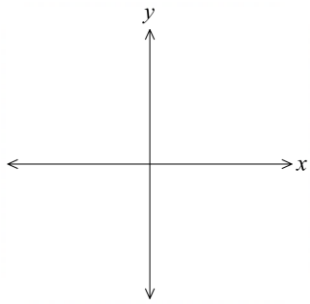
153°

c $3x + 4y = -12$



143°

d $\frac{x}{3} - \frac{y}{2} = 1$



34°

4 What is the angle of inclination of the line $x = -2$?

90°

5 The angle of inclination of a line is 135° and passes through the point $(4,2)$ and $(x,-3)$. Find the value of x.

$m = \tan 135° = -1$

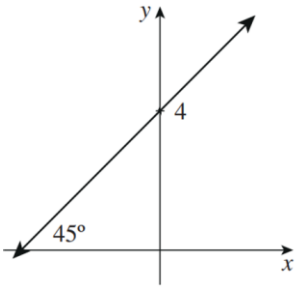
$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3-2}{x-4} = -1$

$\frac{-5}{x-4} = -1$

$x = 9$

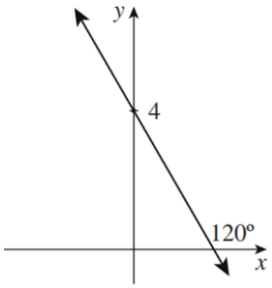
6 Determine the equation of each straight line sketched below in exact general form.

a



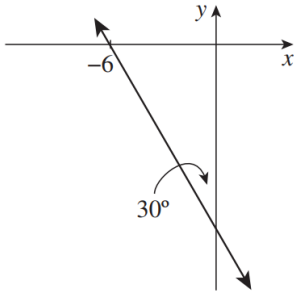
$x - y + 4 = 0$

b



$\sqrt{3}x + y - 4 = 0$

c



$x\sqrt{3} + y + 6\sqrt{3} = 0$

7 In each part, the angle of inclination α and the y-intercept c of a line are given,
Use $m = \tan\alpha$ to find the gradient of each line, then find its equation in exact general form.

a $\alpha = 45°$, $c = (0,3)$

$x - y + 3 = 0$

b $\alpha = 60°$, $c = (0,-1)$

$-\sqrt{3}x + y + 1 = 0$

c $\alpha = 30°$, $c = (0,-2)$

$x - \sqrt{3}y - 2\sqrt{3} = 0$

d $\alpha = 135°$, $c = (0,1)$

$x + y - 1 = 0$

8 In each part, the angle of inclination α and a point A on the line is given,
Use $m = \tan\alpha$ to find the gradient of each line, then find its equation in exact general form.

a $\alpha = 45°$, $A = (1,0)$

$x - y - 1 = 0$

b $\alpha = 120°$, $A = (-1,0)$

$\sqrt{3}x + y + \sqrt{3} = 0$

c $\alpha = 30°$, $A = (4,-3)$

$x - y\sqrt{3} - 4 - 3\sqrt{3} = 0$

d $\alpha = 150°$, $c = (-2,-5)$

$x + \sqrt{3}y + 2 + 5\sqrt{3} = 0$

9 A triangle is formed by the x-axis and the lines $5y = 9x$ and $5y + 9x = 45$.

a Find, correct to the nearest degree, the angles of inclination of the two lines.

b What sort of triangle has been formed?

a) They are about 61° and 119°

b) Since the two interior angles with the x-axis are equal, it is isosceles (a sketch really helps!)

10 Consider the diagram.

a Find the gradient of line AC and hence determine θ, correct to the nearest degree.

b Derive the equation of AC.

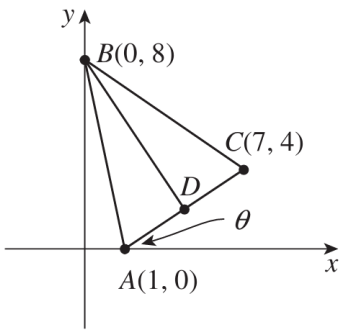
c Find the coordinates of the midpoint D of AC.

d Show that AC is perpendicular to BD.

e What type of triangle is ABC?

f Find the area of this triangle.

g Write the coordinates of a point E such that $ABCE$ is a rhombus.



a) $\frac{2}{3}$, 34°

b) $2x - 3y - 2 = 0$

c) $D(4,2)$

d) $m_{AB} \times m_{BD} = \frac{2}{3} \times -\frac{3}{2} = -1$

e) isosceles

f) $A = \frac{1}{2} \times AC \times BD = \frac{1}{2}(\sqrt{52})(\sqrt{52}) = 26$

g) $E(8, -4)$

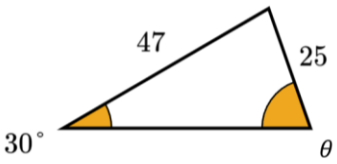
Ambiguous Case of the Sine Rule

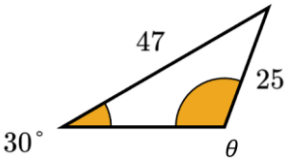
Investigation Ambiguous triangles

Remember the sine rule:

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

Use it to find the unknown angle in the two triangles below.
The first triangle has an acute angle, and the second has an obtuse angle.

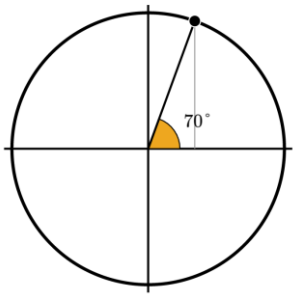




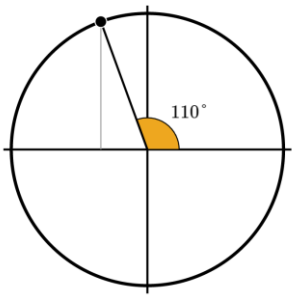
What do you notice about the given lengths and angles of the two triangles?

For some triangles, there are two angle possibilities, we call these triangles **ambiguous.**
This is due to the identity:

$$\sin\theta = \sin(180° - \theta)$$



$\sin(\theta) \approx \quad 0.94$



$\sin(\theta) \approx \quad 0.94$

Investigation Condition for ambiguous triangles

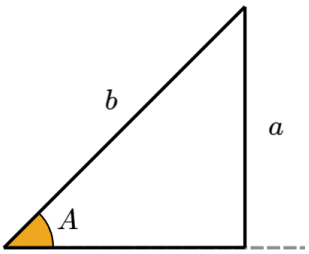
Suppose only side lengths a and b and angle A of a triangle are given.

$\boldsymbol{b \sin a}$ is the **perpendicular height** of the triangle.

https://www.geogebra.org/m/n2wwggdp

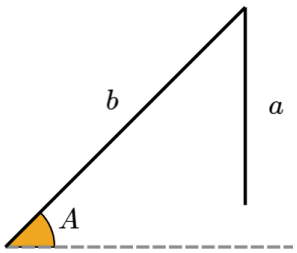
$$\boldsymbol{a = b \sin A}$$

Exactly one triangle is possible



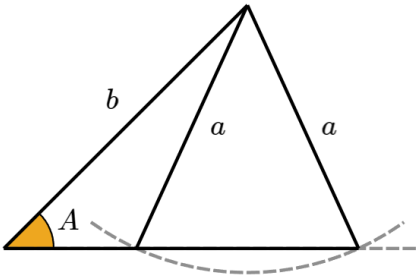
$$\boldsymbol{a < b \sin A}$$

The triangle is not possible.



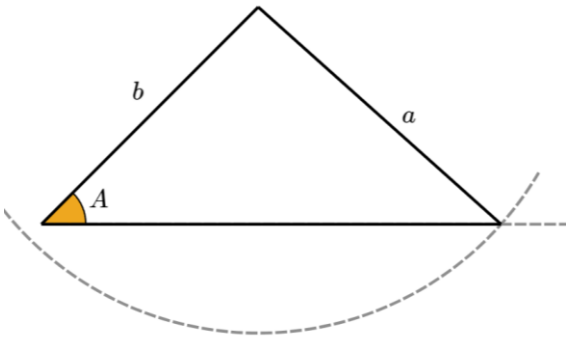
$$\boldsymbol{b \sin A < a < b}$$

Two triangles are possible.



$$\boldsymbol{a > b \sin A}$$

Exactly one triangle is possible.



This is the ambiguous case.

Solving Ambiguous Case Sine Rule Questions

If the question doesn't tell you whether a triangle has an acute or obtuse angle:

1. Use sine rule to find the unknown acute angle θ.
2. Find the obtuse angle $\theta' = 180° - \theta$.
3. Add θ' to the given angle. If the sum $> 180°$, it is not a valid possibility (angle sum of triangle cannot exceed 180°).

⚠ Diagrams are never drawn to scale. In an exam, if you see a sine rule question worth 3 marks or more, it is likely the ambiguous case and marks will be awarded for justifying all possible answers.

Example Calculate angles when the triangle is ambiguous

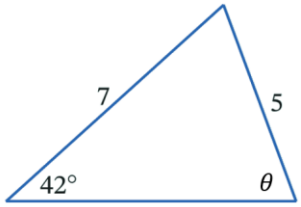
Find the value(s) of θ to the nearest degree.

1.

$$\frac{\sin\theta}{7} = \frac{\sin 42°}{5}$$

$$\sin\theta = \frac{7\sin 42°}{5}$$

$$\theta \approx 70°$$



2.

$$\text{or} \approx 180° - 70°$$

$$\approx 110°$$

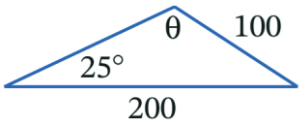
3.

$$110° + 42° = 142° < 180°$$

$$\therefore \theta = 70° \text{ or } 110°$$

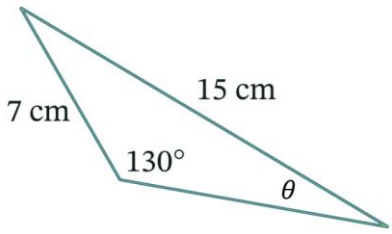
Guided Practice

a Find the value(s) of θ to nearest degree



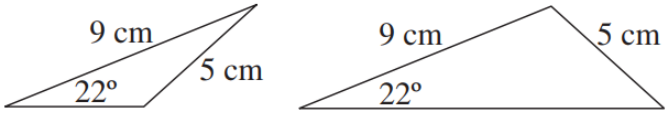
58° or 122°

b Find the value(s) of θ to nearest degree



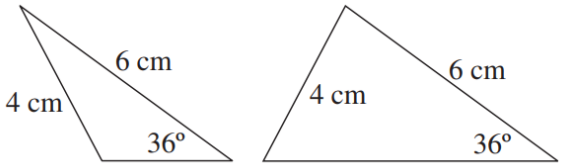
21° only

1 There are two triangles that have sides 9 cm and 5 cm. The angle opposite the 5cm side is 22°. Find, in each case, the size of angle opposite the 9 cm side, correct to the nearest degree.



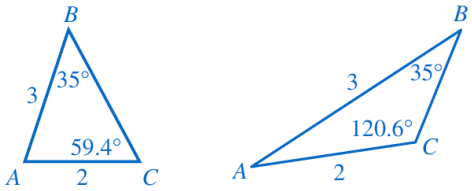
42°, 138°

2 Two triangles are shown, with sides 6 cm and 4 cm. the angle opposite the 4cm is 36°. Find, in each case, the angle opposite the 6cm side, correct to the nearest degree.



62°, 118°

3 A triangle ABC has AB = 3 cm, AC = 2 cm and ∠B = 35°
Find the two possible values of $\angle C$ correct to 1 d.p and sketch the triangles.



59.4° or 120.6°

4 Consider ΔPQR in which $p = 7$ cm, $q = 15$ cm and $\angle P = 25°50'$.

a Find the two possible sizes of $\angle Q$ to the nearest minute then sketch the two triangles.

69°2′ or 110°58′

b For each possible size of $\angle Q$, find side length r in cm.

16.0 cm or 11.0 cm

5 A triangle ABC has AB = 6 m, AC = 10 m and ∠B = 120°.

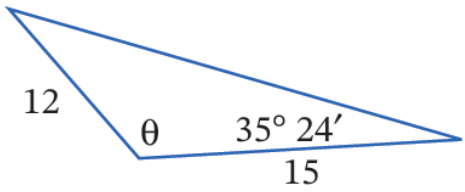
a Find the two possible values of $\angle C$ correct to the nearest degree.

31.3° and 148.7°

b Explain why there is only one solution for $\angle C$.

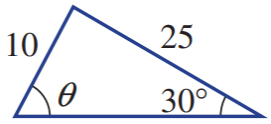
A triangle can only have one obtuse angle as the angle sum is limited to 180°

6 Find θ in this diagram, correct to the nearest minute.



98° 12'

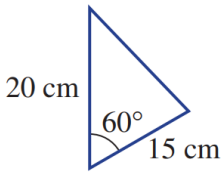
7 Try to find the angle θ in this triangle; explain why it is impossible.



$\theta = \sin^{-1}\left(\frac{25\sin 30°}{10}\right)$, which is impossible bas $\frac{25\sin 30°}{10} > 1$. Therefore, such a triangle is impossible to construct.

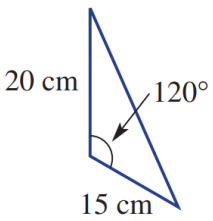
8 Find the area of these two triangles, correct to 2 decimal places.

Triangle 1



129.9 cm²

Triangle 2

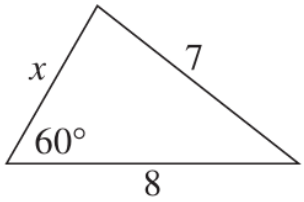


129.9 cm²

What do you notice? How can you explain this?

They are equal as sin 60° and sin 120° are equal.

9 Use the cosine rule to find the two possible values of x in the diagram.



$7^2 = x^2 + 8^2 - 2(x)(8)\cos 60°$

$49 = x^2 + 64 - \frac{16x}{2}$

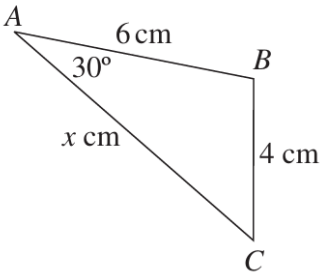
$x^2 - 8x + 15 = 0$

$(x - 3)(x - 5) = 0$

$x = 3$ or 5

10 Consider the diagram.

a Use cosine rule to show $x^2 - 6\sqrt{3}\,x + 20 = 0$



$4^2 = 6^2 + x^2 - 2(6)(x)\cos 30°$

$16 = 36 + x^2 - \frac{12x\sqrt{3}}{2}$

$0 = 20 + x^2 - 6x\sqrt{3}$ as required

b Use the quadratic formula to show AC has length $(3\sqrt{3} + \sqrt{7})$ cm or $(3\sqrt{3} - \sqrt{7})$ cm.

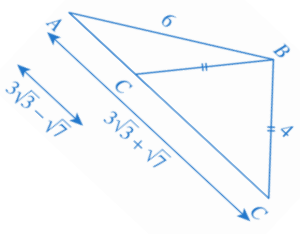
$x = \frac{-(-6\sqrt{3}) \pm \sqrt{(-6\sqrt{3})^2 - 4(1)(20)}}{2}$

$= \frac{6\sqrt{3} \pm \sqrt{28}}{2}$

$= \frac{6\sqrt{3} \pm 2\sqrt{7}}{2}$

$= 3\sqrt{3} \pm \sqrt{7}$

c Indicate on the diagram the other possible position of point C.



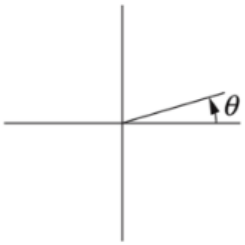
Exam Practice

1 2016 HSC Advanced Band 3

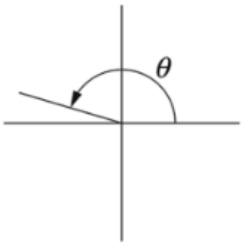
For the angle θ, $\sin\theta = \frac{7}{25}$ and $\cos\theta = -\frac{24}{25}$

Which diagram best shows the angle θ?

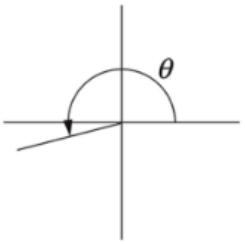
(A)



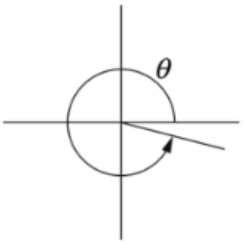
(B)



(C)



(D)

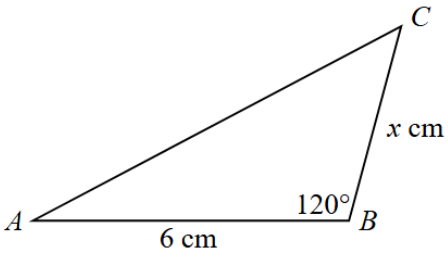


Cos is negative, Sin is positive, so Q2

B

2 HSC Advanced Sample Question Band 4

The triangle shown below has an exact area of $\left(12 - \sqrt{3}\right)$ cm^2.



The exact value of x is:

A. $\frac{24 - 2\sqrt{3}}{3}$

B. $\frac{8\sqrt{3} - 2}{3}$

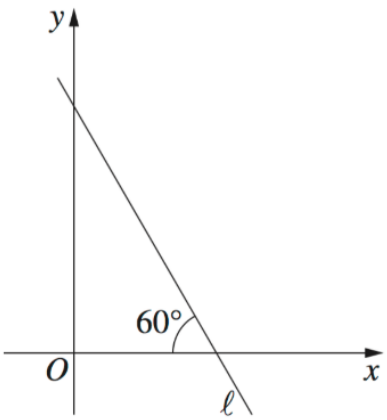
C. $8\sqrt{3} - 1$

D. $12\sqrt{3} + 1$

B

3 2013 HSC Band 4

The diagram shows line l. What is the slope of l?



A. $\sqrt{3}$

B. $-\sqrt{3}$

C. $\frac{1}{\sqrt{3}}$

D. $-\frac{1}{\sqrt{3}}$

$m = \tan 120°$

$= -\tan 60°$

B

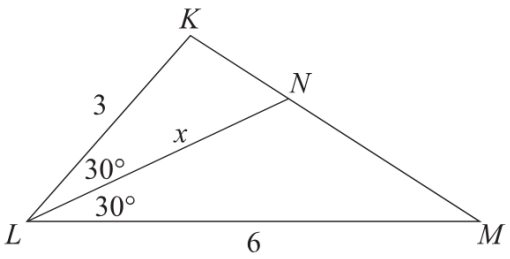
4 HSC Sample Question Band 4

Find all possible lengths of the third side of triangle ABC given $AB = 6.2$ cm, $\angle ABC = 35°$ and $AC = 4.1$ cm.
Answer to 1 d.p.

The third length could be either 7.1 cm or 3.0 cm

5 2018 HSC Advanced Band 5

Find the exact value of the area of ΔKLM and hence find the exact value of x.



$$\text{Area } \Delta KLM = \frac{1}{2}(3)(6)(\sin 60°)$$

$$= \frac{9\sqrt{3}}{2}$$

$$\text{Area } \Delta KLN + \text{Area } \Delta NLM = \text{Area } \Delta KLM$$

$$\frac{3}{4}x + \frac{3}{2}x = \frac{9\sqrt{3}}{2}$$

$$\frac{9x}{4} = \frac{9\sqrt{3}}{2}$$

$$x = 2\sqrt{3}$$

6 HSC Sample Question Band 5

Find the values of θ in $0° \leq \theta \leq 360°$ that satisfy the equation:

$$(\tan\theta - \sqrt{3})(\cos^2\theta - 1) = 0$$

Using null factor law

$$\tan\theta = \sqrt{3}$$

$$\theta = 60°,\ 240°$$

$$\cos^2\theta = 1$$

$$\cos\theta = \pm 1$$

$$\theta = 0,\ 180,\ \text{and } 360°$$

$$\therefore \theta = 0°, 60°, 180°, 240°, 360°$$